

Software Requirements Specification

for

TORBIDD

Software Procurement Intelligence Platform

Version 1.0

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Revision History

Name	Date	Reason for changes	Version
Peraya Pasin Chayakarn	23/08/2026	Initial draft	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for TORBIDD, a web-based platform that discovers, extracts, and structures software-related procurement opportunities published by Thai government agencies. This document translates the project proposal into a formal requirements baseline to guide design, development, testing, and client sign-off. It is intended for the development team, the project client/stakeholders, and QA staff involved in building and validating the system.

1.2 Product Scope

TORBIDD is a centralized website that continuously monitors public procurement announcements beginning with the Bangkok Metropolitan Administration (BMA) and others thai government agencies identifies which listings are software-related projects (as opposed to construction, vehicles, or general goods), and converts unstructured Terms of Reference (TOR) PDF documents into structured, searchable opportunity records. The platform surfaces budget, deadline, and bidder-qualification details that are otherwise buried inside lengthy PDFs, and gives users search, filtering, bookmarking, and alerting tools so they can track and evaluate opportunities without manually reading every announcement.

In scope for the initial release: automated collection of procurement announcements, AI-assisted TOR field extraction, software-project classification, a searchable opportunity dashboard, a procurement detail viewer, a historical budget/price dashboard with outlier analysis, bookmark management, user accounts, an administration panel, and web/email notifications.

Out of scope for the initial release: bid submission or e-procurement transaction processing, payment processing, direct integration with agencies beyond the data sources listed in Section 2.6, and mobile native applications (the initial release is a responsive web application).

1.3 Document Conventions

This document follows standard SRS formatting conventions. Requirements are identified with unique identifiers in the format:

[Feature]-[Number] (e.g. AUTH-1).

Priority levels follow MoSCoW-style labeling: High (must-have for launch), Medium (should-have), Low (nice-to-have, may slip to a later phase).

- “TOR” refers to Terms of Reference, the official document an agency publishes describing a procurement's scope, budget, and requirements.
- “BMA” refers to the Bangkok Metropolitan Administration.
- Monetary values are expressed in Thai Baht (฿) unless noted otherwise.

2. Overall Description

2.1 Product Perspective

TORBIDD is a new, independent, self-contained web-based product designed to improve access to and interaction with public government procurement information. It is not a component of, or replacement for, any existing government e-procurement system.

TORBIDD acts as an aggregation and intelligence layer that collects and processes publicly available procurement announcements and documents published by government sources (Section 2.6). The system then organizes and re-presents this information in a structured, searchable, and user-friendly format to help users discover and understand relevant procurement opportunities.

The system does not modify, submit information to, or authenticate against government agencies' existing procurement systems. It only consumes publicly available data and documents provided by official government sources.

The system architecture consists of

- Web frontend for browsing, searching, and viewing procurement information
- Backend application server for data processing and system logic
- Data collection and integration components for retrieving public procurement information
- Database for storing structured procurement data and related metadata
- Cloud-based file storage for procurement documents
- Search and filtering services for efficient information retrieval
- AI-powered processing and intelligence features for analyzing and organizing procurement information

2.2 Product Functions

At a high level, TORBIDD

- Collects new procurement announcements and TOR documents from monitored government sources on an ongoing basis.
- Extracts structured fields (budget, deadline, scope, bidder qualifications, contact details, etc.) from TOR PDFs using AI-assisted extraction.
- Classifies each opportunity as a software project (or not) and further categorizes it by type/discipline.
- Provides search, filtering, and sorting across the structured opportunity data.
- Displays a detailed view per opportunity and a historical price/budget dashboard with outlier analysis.
- Lets users save, bookmark, and compare opportunities.
- Sends web and email notifications when new or matching opportunities are found.
- Provides administrator tools to monitor data ingestion and manage users and content.

The major functions of the system are grouped as follows

2.2.1 User Authentication and Access Control

- Authenticate users through the supported login mechanism.
- Restrict personalized features such as saved opportunities and notifications to authenticated users.
- Provide administrators with appropriate access to administrative functions.

2.2.2 Procurement Discovery and Search

- Allow users to search procurement opportunities by keywords.
- Provide filters for software category, government agency, budget, and submission deadline.
- Allow users to sort and compare procurement opportunities.
- Display relevant procurement opportunities in a structured format.

2.2.3 Procurement Information and TOR Processing

- Display key procurement information, including title, agency, budget, publication date, deadline, and status.
- Process TOR PDF documents using AI-assisted extraction.
- Extract important information such as project scope, technical requirements, qualifications, and timelines.
- Provide summarized TOR information to help users understand opportunities quickly.
- Provide access to the original procurement announcement and TOR document for verification.

2.2.4 Software Project Classification

- Automatically identify whether a procurement opportunity is software-related.
- Categorize software opportunities into relevant technology or project categories such as AI, ERP, Web Application, Mobile Application, Cloud, and Data Analytics.
- Allow administrators to review and correct classification results.

2.2.5 Procurement Data Collection and Management

- Collect procurement information from monitored government sources.
- Synchronize newly published and updated procurement records.
- Detecting duplicate or outdated records.
- Standardize procurement information from different government sources.
- Allow administrators to monitor data collection and correct inaccurate information.

2.2.6 Historical Procurement and Price Analysis

- Store historical procurement information for reference.
- Allow users to find similar historical software projects.
- Provide historical budget and price comparisons.
- Display budget trends and identify potential budget outliers.

2.2.7 Saved Opportunities and Notifications

- Allow users to save procurement opportunities for later review.
- Notify users about new opportunities matching their interests.
- Notify users when saved opportunities are updated or cancelled.
- Provide reminders for upcoming submission deadlines.

2.3 Stakeholders

Stakeholder	Role	Interest	Influence	Needs / Expectations
Project Sponsor / Customer	Initiates and funds the project	High	High	Wants a centralized platform that helps software companies discover and evaluate software procurement opportunities from BMA and other Thai government agencies efficiently.
Software Companies	Primary users	High	Medium	Wants to quickly find relevant software projects, view bidder qualifications, compare budgets and historical prices, and receive notifications.
Freelance Software Developers	Primary users	High	Low	Need an easy way to identify relevant government software procurement opportunities without reading every TOR.
Business Development / Sales Teams	Primary users	High	Medium	Need to monitor opportunities, estimate project values, compare historical projects, and prioritize potential bids.
Government Procurement Departments	Data providers	Low - Medium	High	Publish procurement announcements and TOR documents that become the source of procurement information through the government API.
Government Project Owners / Agencies	Procurement request owners	Low - Medium	Medium	Create procurement requirements and TOR documents. Their documents affect the quality and completeness of extracted information.
System Administrator	Operates the platform	High	Medium	Manages API synchronization, data updates, users, document processing, classification, and system maintenance.
Development Team	Builds and maintains the system	High	High	Requires clear rules for API integration, document extraction, software-project classification, data normalization, search, and notification logic.
Government Central API / Data Platform	External data source / integration	Medium	High	Provides standardized procurement information from multiple Thai government agencies for automated collection and synchronization.

2.4 User Stories

A. Procurement Discovery and Search

- US1. As a software company, I want to view only software-related government procurement opportunities so that I do not waste time reviewing irrelevant procurement announcements.
- US2. As a freelance software developer, I want to search procurement opportunities by keyword so that I can quickly find projects matching my skills and expertise.
- US3. As a software company, I want to browse procurement opportunities by software category so that I can quickly find projects relevant to my business capabilities.
- US4. As a user, I want to filter procurement opportunities by government agencies so that I can focus on opportunities from agencies that are relevant to me.
- US5. As a business development manager, I want to filter procurement opportunities by budget so that I can prioritize projects within my company's target project value.
- US6. As a business development manager, I want to filter procurement opportunities by submission deadline so that I can prioritize projects that still have enough time for proposal preparation.
- US7. As a user, I want to sort procurement opportunities by publication date so that I can easily identify the newest opportunities.
- US8. As a user, I want to search procurement opportunities using multiple filters at the same time so that I can narrow the results to projects that closely match my requirements.

B. Procurement Information

- US9. As a user, I want to view the basic information of a procurement opportunity so that I can quickly understand what the project is about.
- US10. As a user, I want to view the procurement title, government agency, department, budget, publication date, and deadline so that I can evaluate the opportunity.
- US11. As a software company, I want to view the bidder qualifications so that I can determine whether my company is eligible to bid.
- US12. As a software company, I want to view the required technical qualifications so that I can determine whether my team has the necessary expertise.
- US13. As a user, I want to view the project requirements extracted from the TOR so that I can understand the expected scope of work.
- US14. As a software company, I want to view project timelines and important milestones so that I can estimate whether I have enough resources to complete the project.
- US15. As a user, I want to open the original TOR document so that I can verify the official procurement information.
- US16. As a user, I want to access the original government procurement announcement so that I can verify the information provided by the platform.

C. TOR Processing and Summarization

- US17. As a software company, I want to read a summary of each TOR so that I can understand the project without reading the entire document.

- US18. As a user, I want the system to extract important information from TOR PDFs so that I can quickly find relevant procurement details.
- US19. As a user, I want to view required technologies and technical requirements extracted from the TOR so that I can determine whether the project matches my expertise.
- US20. As a user, I want to view the required qualifications and experience extracted from the TOR so that I can determine whether I am eligible to participate.
- US21. As a system administrator, I want the system to process TOR documents automatically so that procurement information can be extracted without manually entering every field.

D. Software Project Classification

- US22. As a system administrator, I want the system to automatically classify procurement projects as software-related or non-software-related so that irrelevant projects can be filtered out.
- US23. As a system administrator, I want the system to classify software procurements into categories such as AI, ERP, Web Application, Mobile Application, Cloud, Data Analytics, and Information Systems so that users can find relevant projects more easily.
- US24. As a user, I want to see the software category assigned to a procurement so that I can quickly understand the type of project.
- US25. As a system administrator, I want to review incorrectly classified procurement projects so that classification accuracy can be improved.

E. Government Central API Integration

- US26. As a system administrator, I want the system to regularly collect procurement announcements from the Thai government central API so that users always have access to the latest opportunities.
- US27. As a system administrator, I want the system to retrieve procurement information from multiple Thai government agencies through the central API so that the platform provides a broader range of opportunities.
- US28. As a system administrator, I want the system to automatically synchronize newly published procurement records so that new opportunities appear on the platform without manual data entry.
- US29. As a system administrator, I want the system to synchronize updated procurement information so that changes made by government agencies are reflected on the platform.
- US30. As a system administrator, I want the system to detect cancelled or closed procurement opportunities so that users do not spend time preparing bids for inactive projects.
- US31. As a system administrator, I want the system to record the date and time of the latest API synchronization so that I can monitor data freshness.
- US32. As a system administrator, I want the system to detect duplicate procurement records so that the same opportunity is not displayed multiple times.
- US33. As a system administrator, I want procurement data from different government agencies to be converted into a standardized format so that users can search and compare information consistently.

- US34. As a system administrator, I want the system to identify the source government agency and department for each procurement so that users can determine where the opportunity originated.
- US35. As a system administrator, I want the system to record API synchronization errors so that failed data collection can be identified and resolved.
- US36. As a system administrator, I want the system to retry failed API synchronization processes so that temporary connection or service failures do not cause permanent data gaps.

F. User Accounts and Personalization

- US37. As a user, I want to create an account so that I can access personalized features.
- US38. As a user, I want to log in securely so that my saved opportunities, preferences, and notifications are associated with my account.
- US39. As a user, I want to manage my profile and areas of expertise so that the system can recommend relevant procurement opportunities.
- US40. As a user, I want to save procurement opportunities so that I can review them later before deciding whether to bid.

G. Notifications

- US41. As a software company, I want to receive notifications when a new procurement matches my interests so that I do not miss potential business opportunities.
- US42. As a user, I want to receive notifications when a saved procurement is updated so that I know when important information has changed.
- US43. As a user, I want to receive reminders about upcoming procurement deadlines so that I can submit my proposal on time.

H. Historical Procurement and Price Comparison

- US44. As a business development manager, I want to compare the announced budget with historical government software projects so that I can determine whether the estimated budget is reasonable or an outlier.
- US45. As a software company, I want to view similar historical software procurements so that I can estimate the expected project scope and pricing.
- US46. As a business development manager, I want to compare project budgets across different government agencies so that I can understand differences in government software procurement values.

I. Administrator and Data Management

- US47. As a system administrator, I want to manually correct extracted procurement information when necessary so that incorrect data does not affect users. 
- US48. As a system administrator, I want to review failed TOR document processing so that documents with extraction problems can be handled. 
- US49. As a system administrator, I want to monitor data quality and completeness so that missing or incorrect procurement information can be identified. 

2.5 User Classes and Characteristics

Primary User Class: Software Companies & Studios

- **Frequency of use:** Regular to frequent
- **Technical expertise:** Moderate to high
- **Primary functions used:**
 - Searching and filtering public and private opportunities
 - Monitoring newly available opportunities
 - Setting up recurring monitoring activities
 - Comparing opportunities based on requirements, budget, and timeline
 - Reviewing historical pricing and opportunity data
- **Characteristics:** Organizational users who need continuous monitoring of relevant opportunities rather than performing one-off searches.
- **Priority:** High - this is one of the main user groups of the system.

Secondary User Class: Freelance Developers

- **Frequency of use:** Occasional to regular
- **Technical expertise:** High
- **Primary functions used:**
 - Searching and filtering opportunities
 - Filtering opportunities based on technical requirements and skills
 - Comparing opportunities based on suitability and budget
 - Reviewing opportunity details and requirements
- **Characteristics:** Individual developers who are highly sensitive to skill requirements, pricing, and opportunity relevance and are likely to make extensive use of filtering features.
- **Priority:** High - this user group relies heavily on accurate filtering and opportunity matching.

Secondary User Class: System Integrators & IT Consultants

- **Frequency of use:** Regular
- **Technical expertise:** Moderate to high
- **Primary functions used:**
 - Searching and comparing public and private opportunities
 - Reviewing opportunity requirements and budgets
 - Analyzing historical pricing data
 - Comparing opportunities to identify suitable projects
- **Characteristics:** Organizational or individual professional users who need to evaluate different types of opportunities and use historical data to support business and pricing decisions.
- **Priority:** Medium - supports opportunity comparison and data-driven decision-making.

Secondary User Class: Business Development / Sales Teams

- **Frequency of use:** Regular
- **Technical expertise:** Low to moderate
- **Primary functions used:**
 - Searching and filtering opportunities
 - Reviewing summarized budgets and timelines
 - Comparing multiple opportunities
 - Reviewing simplified opportunity information instead of raw TOR text
- **Characteristics:** Business-focused users who may have limited technical knowledge and therefore require clear summaries, comparisons, budgets, and timelines rather than detailed technical documents.
- **Priority:** Medium - supports business development and sales activities.

Administrative User Class: System Administrators

- **Frequency of use:** Frequent
- **Technical expertise:** High
- **Primary functions used:**
 - Monitoring data ingestion
 - Correcting misclassified or inaccurate data
 - Managing user accounts
 - Maintaining data quality and system operations
- **Characteristics:** Internal system users responsible for ensuring that collected opportunity data is accurate, correctly classified, and available to other users.
- **Priority:** High - required for maintaining the reliability and operational quality of the system.

2.6 Operating Environment

The TORBIDD system will operate in the following environment

- **Client-side:** Modern desktop and mobile web browsers, including Chrome, Firefox, and other commonly supported browsers. The application will use a responsive layout to support different screen sizes.
- **Server-side:** Cloud-based production environment with the backend deployed using Docker containers.
- **Database:** MongoDB Atlas will be used to store structured opportunity data, including information extracted from tender and opportunity documents.
- **AI and document processing:** Vertex AI will be used for TOR PDF ingestion, document processing, and information extraction.
- **External data sources:** The system will continuously poll or receive data from public government sources, including:
 - Bangkok Metropolitan Administration e-Procurement: egp2.bangkok.go.th

- Electronic Government Procurement: www.gprocurement.go.th
- Open Government Data of Thailand: data.go.th
- **Network:** Internet connectivity is required for accessing the web application, government data sources, MongoDB Atlas, Vertex AI, and other required services.
- **Deployment:** The production system will be containerized using Docker to provide a consistent and maintainable deployment environment.

2.7 Design and Implementation Constraints

The design and implementation of TORBIDD are constrained by the following project requirements and limitations

- **CON-1:** The total project budget is limited to ฿1,000,000, as specified in the project proposal.
- **CON-2:** The system must be delivered within a **12-week development period** following requirements sign-off.
- **CON-3:** The backend system shall use **Node and Next.js** as the primary web application framework.
- **CON-4:** The system shall use **MongoDB Atlas** for storing structured procurement data and related metadata.
- **CON-5:** The system shall use **Vertex AI** for AI-assisted TOR document extraction, summarization, and procurement classification.
- **CON-6:** The production application shall be deployed using **Docker** within a cloud-based hosting environment.
- **CON-7:** The system shall provide a **responsive web interface** that supports modern desktop and mobile web browsers.
- **CON-8:** The system shall consume publicly available procurement information and shall not modify, submit data to, or directly manage the source government procurement systems.
- **CON-9:** The system shall support **Thai TOR content** for document processing and information extraction.
- **CON-10:** The system shall include administrative review and correction capabilities because AI-based document extraction and classification cannot be assumed to be completely accurate.
- **CON-11:** The system shall be developed within the resources and staffing limitations defined by the project team and schedule.

2.8 Assumptions and Dependencies

2.8.1 Assumptions

The following assumptions are made during the specification and development of TORBIDD. Changes to these assumptions may require corresponding changes to the system design or requirements.

- The monitored government sources will continue to publish procurement announcements and TOR documents publicly through web pages, downloadable documents, or public APIs.
- TOR documents are expected to contain Thai text. The document-processing pipeline is therefore assumed to support Thai content.
- Procurement information contained in government announcements and TOR documents is assumed to be sufficiently complete and accurate for extraction and presentation by the

system.

- Software-related procurement opportunities can be identified and categorized with reasonable accuracy using automated classification, supported by administrator review and correction when necessary.
- Users have a valid email address for receiving system notifications.
- Users have reliable Internet access to use the web application and receive updated procurement information.

2.8.2 Dependencies

TORBIDD depends on the following external services and infrastructure:

- **Government Data Sources:** The system depends on the availability and accessibility of monitored government websites and public APIs, including e-procurement and open-data sources.
- **Vertex AI:** The system depends on Vertex AI for AI-assisted TOR document extraction, summarization, and procurement classification. Changes to its availability, API, pricing, or supported capabilities may affect system functionality.
- **MongoDB Atlas:** The system depends on MongoDB Atlas for storing structured procurement information, user-related data, and metadata.
- **Production Hosting Infrastructure:** The system depends on a cloud-based hosting environment capable of running the application's Docker containers and supporting the required backend services.
- **Email Notification Service:** The notification functionality depends on an available email delivery service for sending procurement alerts and reminders.

2.8.3 Impact of Assumption Changes

- If a monitored government source changes its website structure, document format, API, access policy, or removes public access, the corresponding data-collection component may require modification, and data from the affected source may temporarily become unavailable.
- If TOR documents cannot be processed accurately due to changes in document structure, language, or extraction quality, administrator review and manual correction will be required to maintain data quality.
- If Vertex AI becomes unavailable, changes its API, or no longer provides the required capabilities, AI-based TOR extraction, summarization, and classification may be partially or fully unavailable until an alternative solution is implemented.
- If MongoDB Atlas or the production hosting infrastructure becomes unavailable, users may be unable to access or update procurement information until the affected service is restored.
- If the email notification service becomes unavailable, procurement alerts and reminders may not be delivered, while other core system functions may continue to operate.

2.9 Requirements Analysis Notes

This section documents unresolved issues, ambiguities, and justified assumptions identified during the requirements analysis process. These notes identify areas where the proposal or client requirements were incomplete, high-level, or required interpretation.

2.9.1 Open Issues

The following issues may require clarification during further requirements analysis or future development

- **OI-1: Government Source Scope**, The exact list of government agencies to be monitored beyond the sources identified in Section 2.6 is not fully finalized. The proposal indicates coverage of up to five government agencies, subject to confirmation.
- **OI-2: Classification Correction Feedback**, It has not been determined whether administrator corrections to AI-generated classifications and extracted information should be used to improve future AI processing or remain as manual overrides.
- **OI-3: Procurement Data Retention**, The retention policy for expired, cancelled, or closed procurement opportunities has not been specified. It remains to be determined whether such records should be archived or permanently deleted.

2.9.2 Unclear or Ambiguous Requirements

The following requirements require further clarification or a defined implementation approach

- **UR-1: Budget Outlier Analysis**, The proposal specifies budget outlier analysis but does not define the statistical method or threshold to be used. A specific method, such as percentile-based thresholds or standard-deviation analysis, should be defined before implementation.
- **UR-2: User Profile Scope**, The exact scope of the user profile feature is not fully defined. It is unclear whether the profile should include only basic user information or additional preferences such as technology interests, saved searches, or organization information.

2.9.3 Justified Assumptions

Where the proposal does not specify sufficient implementation detail, the following assumptions are made to allow the requirements to be defined and tested

- **JA-1: Notification Channels**, Notifications will support both in-app/web notifications and email notifications.
- **JA-2: Administrator Functions**, The administration panel will provide, at minimum, user management, procurement-data review, classification correction, and data-quality management functions.
- **JA-3: Saved Opportunities**, Users will be able to save procurement opportunities to a personal list and later remove them from that list.
- **JA-4: Historical Data Retention**, Historical procurement records will be retained for analysis and comparison unless a future data-retention policy specifies otherwise.
- **JA-5: AI Processing Review**, AI-extracted information and classifications may require administrator review or correction when the system identifies errors or insufficient confidence.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a responsive web-based user interface accessible through modern desktop and mobile web browsers.

- **UI-1**: The system shall provide a user authentication interface for account login.
- **UI-2**: The system shall provide a procurement discovery interface for browsing and searching

available opportunities.

- **UI-3:** The system shall provide search and filtering controls for criteria such as keywords, software category, government agency, budget, and submission deadline.
- **UI-4:** The system shall display procurement opportunities in a structured list or search-results interface.
- **UI-5:** The system shall provide a detailed procurement view containing information such as title, agency, department, budget, publication date, deadline, status, qualifications, technical requirements, and project scope.
- **UI-6:** The system shall provide a summarized view of information extracted from TOR documents.
- **UI-7:** The system shall provide access to the original procurement announcement and TOR document where available.
- **UI-8:** The system shall allow users to save and remove procurement opportunities from their personal saved list.
- **UI-9:** The system shall provide notification interfaces for new, updated, matching, or upcoming procurement opportunities.
- **UI-10:** The system shall provide an administrative interface for monitoring data collection, reviewing extracted information, and correcting classification or data errors.

Detailed UI layouts, visual design, and styling specifications may be documented separately in the UI design specification.

3.2 Software Interfaces

Interface	Purpose
BMA e-Procurement (egp2.bangkok.go.th)	Source of procurement announcements and TOR documents.
Electronic Government Procurement (gprocurement.go.th)	Additional source of government procurement information.
Open Government Data (data.go.th)	Supplementary source of public government and procurement-related data.
Vertex AI	AI-assisted TOR document extraction, summarization, and software-project classification.
MongoDB Atlas	Storage of structured procurement data, user information, saved opportunities, and related metadata.
Email Delivery Service	Delivery of procurement notifications and alerts; the specific provider is subject to confirmation.

4. System Features

4.1 User Authentication and Access Control

4.1.1 Description and Priority

This feature allows users to securely authenticate using Google Sign-In before accessing any system functionality. All features of the system are restricted to authenticated users.

Priority: Medium

4.1.2 Response Sequence

- Users access the system.
- The system displays the Google Sign-In option.
- User authenticates using Google OAuth.
- The system verifies authentication and grants access to the application.
- If authentication fails, the system displays an error message and denies access.

4.1.3 Functional Requirements

- AUTH-1: The system shall allow users to authenticate using Google Sign-In.
- AUTH-2: The system shall restrict access to all system features to authenticated users only.
- AUTH-3: The system shall prompt unauthenticated visitors to sign in before accessing content.

4.2 Procurement Opportunity Dashboard

4.2.1 Description and Priority

The dashboard serves as the homepage for users, displaying the latest software-related procurement opportunities published by BMA.

Priority: High

4.2.2 Response Sequence

- The user selects the opportunities on the side bar.
- The system displays a procurement opportunities dashboard.

4.2.3 Functional Requirements

- POD-1: The system shall display active software-related procurement opportunities from BMA.

- POD-2: The system shall display the following details for each procurement item on the dashboard: Project Title, Issuing Department, Estimated Budget, and Submission Deadline.
- POD-3: The system shall allow users to click on any procurement listing to navigate to its full details page.

4.3 Smart Search & Filtering

4.3.1 Description and Priority

Users can efficiently search and filter procurement opportunities to find projects relevant to their expertise.

Priority: High

4.3.2 Response Sequence

- The user input the relevant information.
- The system displays all procurement that match the user's search and filter.

4.3.3 Functional Requirements

- SSF-1: The system shall allow users to search procurement opportunities using keywords.
- SSF-2: The system shall allow users to filter procurement opportunities by Issuing Department.
- SSF-3: The system shall allow users to filter procurement opportunities by Estimated Budget Range.
- SSF-4: The system shall allow users to filter procurement opportunities by Submission Deadline.
- SSF-5: The system shall allow users to filter procurement opportunities by Project Category.
- SSF-6: The system shall update the dashboard feed dynamically based on the applied search and filter criteria.

4.4 TOR Information Extraction

4.4.1 Description and Priority

The system automatically extracts key information from TOR PDF documents and presents it in a structured format.

Priority: High

4.4.2 Response Sequence

- User accesses their interest opportunities.
- The system retrieves and extracts the information from TOR PDF documents.
- The system displays the information in a structured format.

4.4.3 Functional Requirements

- TIE-1: The system shall automatically extract important information (Project Title, Department, Budget, Project Scope, Submission Deadline, and Bidder Qualifications) from TOR PDF documents.
- TIE-2: The system shall store the extracted data in the database linked to the respective procurement record.

4.5 Software Project Classification

4.5.1 Description and Priority

The system automatically identifies whether a procurement project is software-related.

Priority: High

4.5.2 Response Sequence

- The system searches for TOR PDFs.
- The system identifies whether that TOR PDF is software related or not.

4.5.3 Functional Requirements

- SPC-1: The system shall automatically classify whether a procurement is software-related.
- SPC-2: The system shall classify a project as software-related if it matches one or more of the following criteria: Website development, Mobile application, Information systems, AI solutions, Database systems, or Software maintenance.

4.6 Procurement Detail Viewer

4.6.1 Description and Priority

Users can view complete procurement information extracted from the TOR.

Priority: High

4.6.2 Response Sequence

- The user selects a procurement opportunity that they are interested in.

- The system displays all of the opportunity details in a structured format.

4.6.3 Functional Requirements

- PDV-1: The system shall display detailed information for every procurement opportunity.
- PDV-2: The system shall display the structured data extracted from the TOR, including: Project Title, Budget, Department, Project Description, Scope of Work, Bidder Qualifications, and Submission Deadline.
- PDV-3: The system shall provide a link or embedded viewer for users to access and download the original TOR PDF document.

4.7 Opportunity Notification

4.7.1 Description and Priority

The system notifies users when new procurement opportunities match their interests.

Priority: Medium

4.7.2 Response Sequence

- The user selects their interest tags.
- The system finds new TOR PDFs.
- The system matches the new opportunity with the interest tags from the user.
- The system notifies the user if the new opportunity matches with the user's interest.

4.7.3 Functional Requirements

- ON-1: The system shall notify users when a new procurement matches their interests.
- ON-2: The system shall allow users to define, update, or toggle their notification criteria and delivery preferences.

4.8 Historical Price Dashboard

4.8.1 Description and Priority

Displays historical software procurement data for comparison.

Priority: High

4.8.2 Response Sequence

- The user selects the price analysis on the side bar.
- The system displays historical software procurement data.

4.8.3 Functional Requirements

- HPD-1: The system shall display historical software procurement projects for comparison.
- HPD-2: The system shall allow users to filter the historical data by Year.
- HPD-3: The system shall allow users to filter the historical data by Department.
- HPD-4: The system shall allow users to filter the historical data by Project Category.

4.9 Budget Outlier Analysis

4.9.1 Description and Priority

The system analyzes whether a procurement budget appears significantly different from similar historical software projects.

Priority: High

4.9.2 Response Sequence

- The user selects the price analysis on the side bar.
- The system displays historical software procurement data.

4.9.3 Functional Requirements

- BOA-1: The system shall compare the current procurement budget with historical software procurements.
- BOA-2: The system shall generate and display a visual warning if the current project's budget is significantly higher or lower than the historical average.

4.10 Bookmark Management

4.10.1 Description and Priority

Users can save procurement opportunities for later review.

Priority: Low

4.10.2 Response Sequence

- The user clicks on the save opportunity button.
- The system saves that opportunity to the "saved item" section.
- The user selects the Saved Items on the side bar to view the procurement.

4.10.3 Functional Requirements

- BM-1: The system shall allow users to bookmark procurement opportunities.
- BM-2: The system shall allow users to remove a bookmark from a previously saved opportunity.

4.11 User Profile

4.11.1 Description and Priority

Users manage their account information and notification preferences.

Priority: Medium

4.11.2 Response Sequence

- The user selects the user icon.
- The system displays that user's profile.
- The user can update their profile and manage notifications

4.11.3 Functional Requirements

- UP-1: The system shall allow users to manage their account.
- UP-2: The system shall allow users to configure their notification preferences.

4.12 Administration Panel

4.12.1 Description and Priority

Allows administrators to manage the platform.

Priority: Medium

4.12.2 Response Sequence

- The admin login as an admin.
- The system displays the admin panel.

4.12.3 Functional Requirements

- AP-1: The system shall provide a secure, role-restricted Administration Panel accessible only to authorized admin users.
- AP-2: The system shall allow admins to monitor the status of automatic procurement data ingestion.
- AP-3: The system shall allow admins to view, edit, or manually correct data extracted from TOR documents.
- AP-4: The system shall allow admins to override or correct the automated Software Project Classification.
- AP-5: The system shall allow admins to view system processing logs to troubleshoot extraction or classification errors.
- AP-6: The system shall allow admins to manage user accounts, including viewing, suspending, or deleting users.

5. Use Case Model

5.1 Actors

1. Registered User

- The primary user of TORBIDD.
- Represents software companies, software studios, freelance developers, IT consultants, system integrators, and business development or sales personnel.
- Can search and filter procurement opportunities, view TOR details, bookmark and compare opportunities, configure notifications, and access historical budget information.

2. Guest / Unauthenticated Visitor

- An individual who accesses TORBIDD without signing in.
- Can browse, search, and view publicly available procurement opportunities.
- Must register or sign in before using personalized functions such as bookmarking, comparison, and notifications.

3. System Administrator

- An internal staff member responsible for maintaining the quality and operation of TORBIDD.
- Reviews low-confidence extraction results, corrects extracted fields, changes project classifications, rejects duplicate or irrelevant records, manages users, and monitors data collection.

4. Data Collection Service

- An automated background service that periodically checks monitored government procurement sources.
- Detects new announcements, downloads TOR documents, invokes the extraction and classification process, and stores structured opportunity records.

5. Government Procurement Data Source

- External government websites that publish procurement announcements and TOR documents.
- Initial sources include BMA e-Procurement, gprocurement.go.th, data.go.th, OCSC, and the Ministry of Labour.

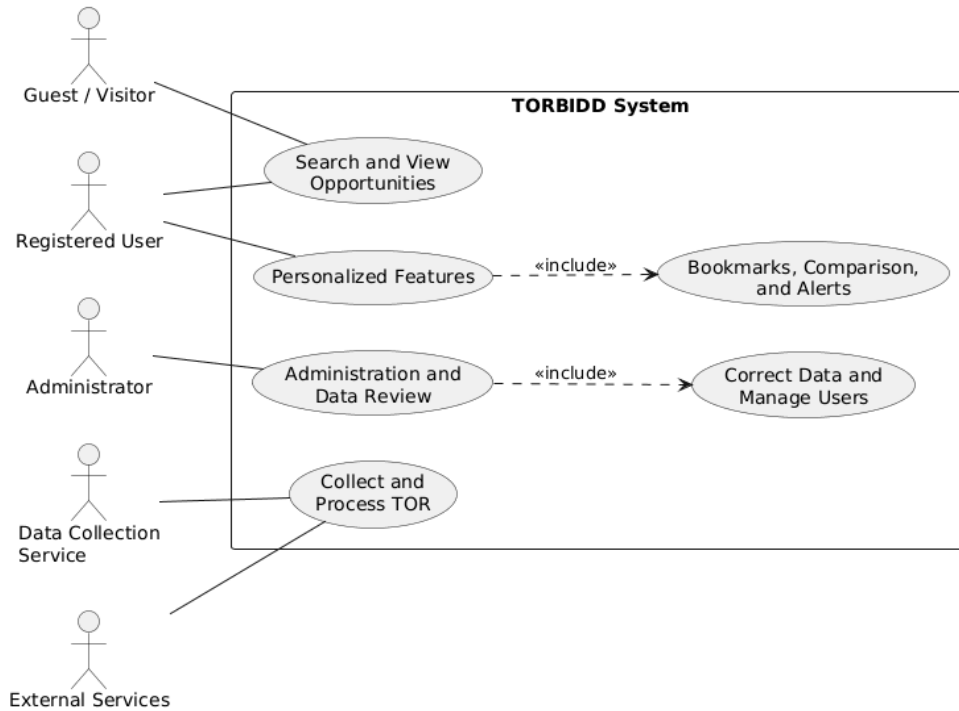
6. Third-Party AI Service

- An external AI service, such as Vertex AI, used to process Thai TOR documents.
- Extracts structured fields and assists in classifying each procurement as a software or non-software project.

7. Email Delivery Service

- An external service responsible for sending account-related emails and procurement opportunity notifications.
- The specific email provider will be determined during system design.

5.2 Use case diagram



5.3 Use case descriptions

5.3.1 Use Case List

ID	Use Case Name
UC-1	Sign In with Google OAuth
UC-2	Search and Filter Opportunities
UC-3	View Opportunity Details
UC-4	Bookmark Opportunity
UC-5	Compare Opportunities
UC-6	View Historical Budget and Outlier Analysis
UC-7	Configure and Receive Opportunity Alerts
UC-8	Collect Procurement Announcements
UC-9	Extract and Classify TOR Information
UC-10	Review and Correct Opportunity Information
UC-11	Manage Users
UC-12	Manage User Profile and Preferences

Note: The level of detail in user flow diagrams does not necessarily correspond to the textual use case type and is intended to improve understanding of system interactions.

UC-1: Sign In with Google OAuth

Use Case Type: Casual

Primary Actor: Guest / Unauthenticated Visitor

Supporting Actor: Google OAuth Authentication Service

Description:

A visitor signs in to TORBIDD using their Google account to access personalized features such as bookmarks, comparisons, saved opportunities, and notifications. If the Google account is not yet associated with a TORBIDD account, the system creates a new TORBIDD account using the information provided by Google.

Preconditions:

- The visitor can access the TORBIDD website.
- The visitor is not currently authenticated.
- The visitor has a valid Google account.

Main Flow:

1. The visitor opens the TORBIDD sign-in page.
2. The system displays the **Sign in with Google** button.
3. The visitor selects **Sign in with Google**.
4. The system redirects the visitor to Google's OAuth authentication page.
5. The visitor authenticates with their Google account and grants the required permissions.
6. Google returns an OAuth authorization result to TORBIDD.
7. The system validates the OAuth result and retrieves the required Google account information.
8. The system checks whether the Google account is already associated with a TORBIDD account.
9. If an existing account is found, the system authenticates the visitor.
10. If no existing account is found, the system creates a new TORBIDD account using the available Google account information.
11. The system creates an authenticated session.
12. The system redirects the visitor to the procurement opportunity dashboard.
13. Personalized features such as bookmarks, comparisons, saved opportunities, and notifications become available.

Alternative Flows:

- **A1: Google Authentication Cancelled**
If the visitor cancels the Google authentication process, the system returns the visitor to the TORBIDD sign-in page and displays an appropriate message.
- **A2: Google Authentication Fails**
If Google authentication fails, the system does not create an authenticated session and displays an authentication error.
- **A3: OAuth Authorization Result Invalid**
If the OAuth authorization result is invalid, expired, or cannot be verified, the system denies authentication and displays an appropriate error message.

- **A4: Existing TORBIDD Account**

If the visitor's Google account is already associated with a TORBIDD account, the system signs the visitor into the existing account rather than creating a new account.

- **A5: Account Suspended**

If the Google account is associated with a suspended TORBIDD account, the system denies access and displays an appropriate message.

- **A6: Required Account Information Missing**

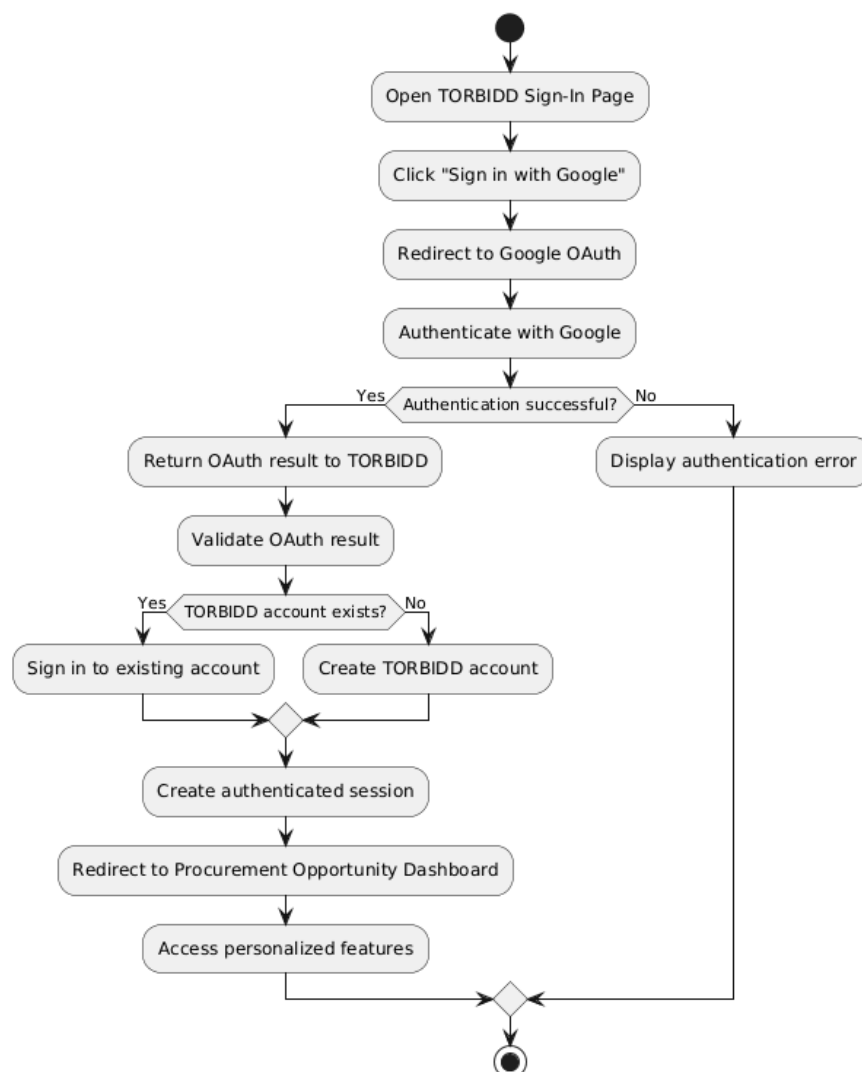
If required information cannot be obtained from Google, the system asks the visitor to provide the missing information before completing account creation.

Postconditions:

- The visitor is successfully authenticated through Google OAuth, or a new TORBIDD account is created.
- An authenticated TORBIDD session is established.
- Personalized features become available.
- The visitor is redirected to the procurement opportunity dashboard.

User Flow Diagram (Sign In with Google OAuth):

UC-1: Sign In with Google OAuth



UC-2: Search and Filter Opportunities

Use Case Type: Casual

Primary Actors: Guest, Registered User

Description:

The actor searches for software-related government procurement opportunities using keywords and structured filters.

Preconditions:

- The system contains at least one published procurement opportunity.
- The actor has opened the opportunity dashboard.

Main Flow:

1. The actor opens the procurement opportunity dashboard.
2. The system displays a paginated list of current software-related opportunities.
3. Each opportunity displays its title, issuing agency, budget, submission deadline, and category.
4. The actor enters one or more keywords.
5. The actor optionally applies filters such as:
 - Budget range
 - Issuing agency
 - Submission deadline
 - Software-project category
6. The actor optionally selects a sorting option.
7. The system queries the structured opportunity database.
8. The system displays matching opportunities.
9. The actor selects an opportunity to view its details.

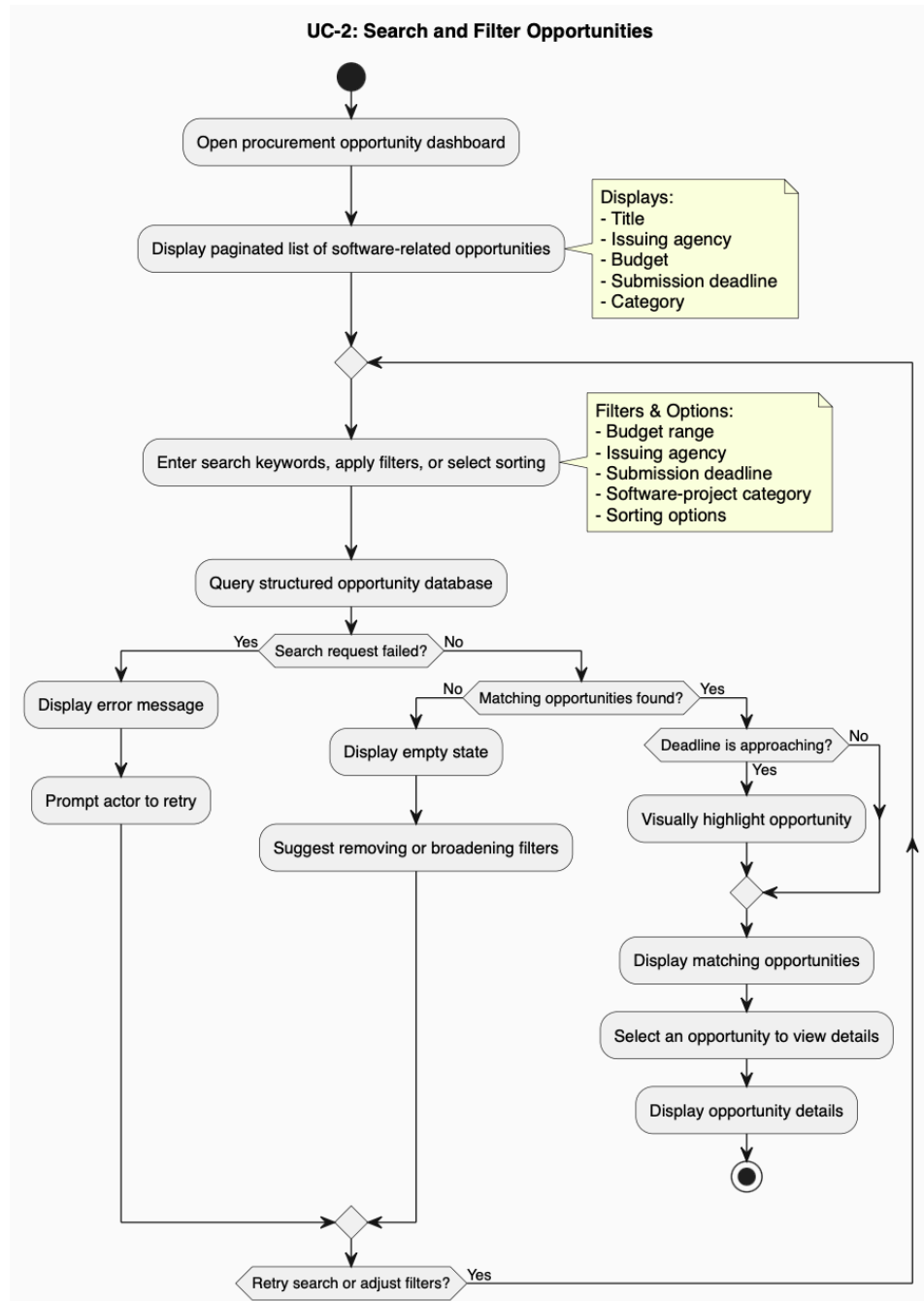
Alternative Flows:

- If no opportunities match, the system displays an empty state.
- The system suggests removing or broadening one or more filters.
- If the search request fails, the system displays an error and allows the actor to retry.
- If a deadline is approaching, the system visually highlights the opportunity.

Postconditions:

- Matching procurement opportunities are displayed.
- The selected search and filter criteria remain active until cleared or changed.

User Flow Diagram (Search and Filter Opportunities):



UC-3: View Opportunity Details

Use Case Type: Casual

Primary Actors: Guest, Registered User

Description:

The actor views structured information extracted from a selected TOR document without having to manually read the complete PDF.

Preconditions:

- The selected opportunity has been collected and classified as software-related.
- The opportunity is published on the dashboard.

Main Flow:

1. The actor selects an opportunity from the dashboard or search results.
2. The system retrieves the structured opportunity record.
3. The system displays:
 - Project title
 - Issuing government agency
 - Procurement budget
 - Submission deadline
 - Project scope
 - Expected deliverables
 - Bidder qualifications
 - Contact information
 - Project category
 - Extraction or classification confidence
4. The system displays a link to the original announcement or TOR PDF.
5. The actor reviews the information.
6. A registered user may proceed to bookmark, compare, or configure an alert for the opportunity.

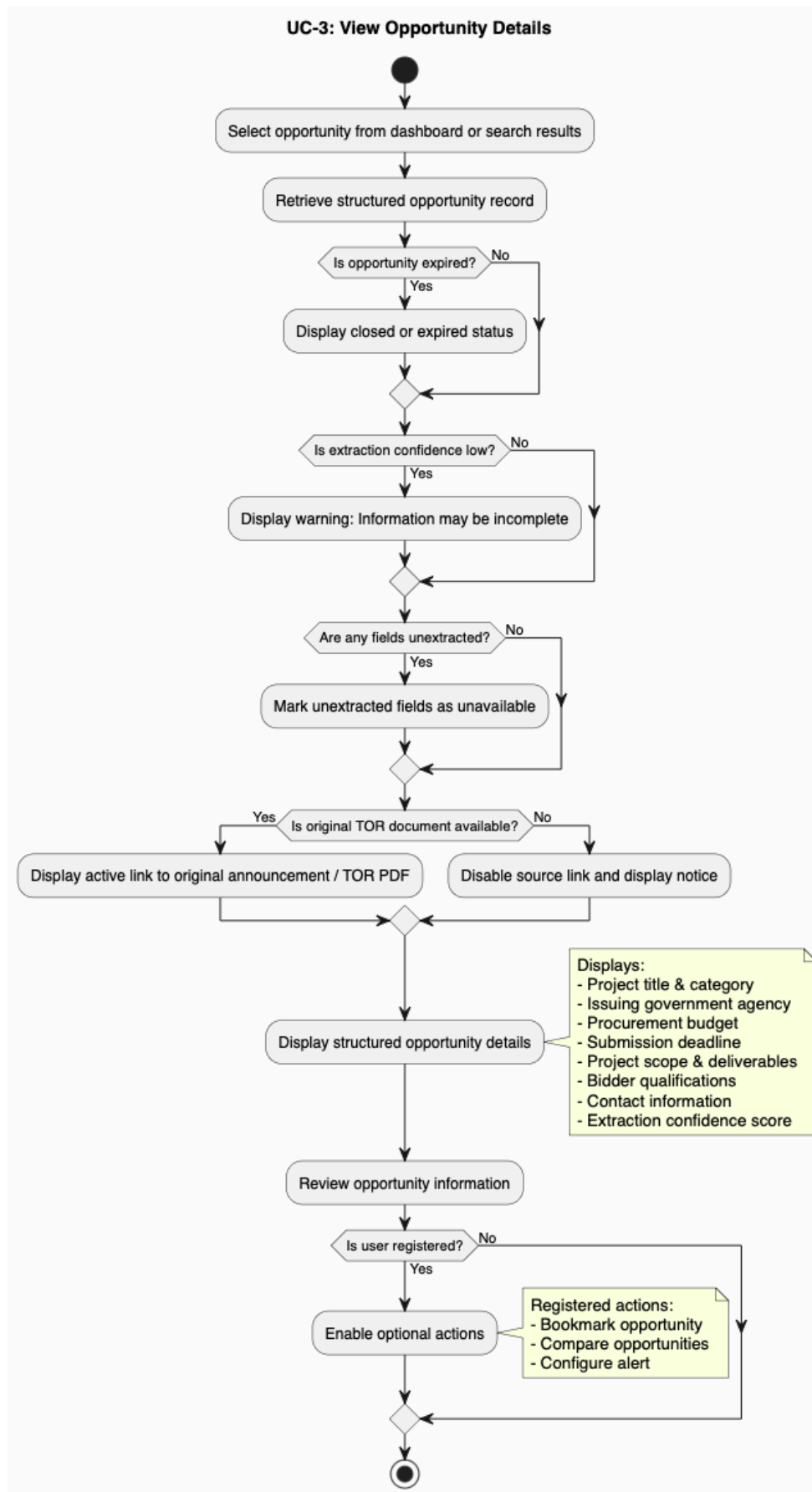
Alternative Flows:

- If extraction confidence is low, the system displays a warning that some information may be incomplete.
- If a field could not be extracted, the system marks it as unavailable.
- If the original TOR document is unavailable, the system disables the source link and displays a notice.
- If the opportunity has expired, the system displays its closed or expired status.

Postconditions:

- The actor has viewed the structured opportunity information.
- No procurement information is modified.

User Flow Diagram (View Opportunity Details):



UC-4: Bookmark Opportunity

Primary Actor: Registered User

Description:

A registered user saves an opportunity to a personal bookmark list for later review.

Preconditions:

- The user is authenticated.
- The selected opportunity is available in the system.

Main Flow:

1. The user opens an opportunity from the dashboard, search results, or detail page.
2. The user selects the bookmark option.
3. The system checks whether the opportunity is already bookmarked.
4. The system adds the opportunity to the user's saved opportunities.
5. The system displays bookmark confirmation.
6. The user can later access the opportunity from the personal saved-opportunities page.

Alternative Flows:

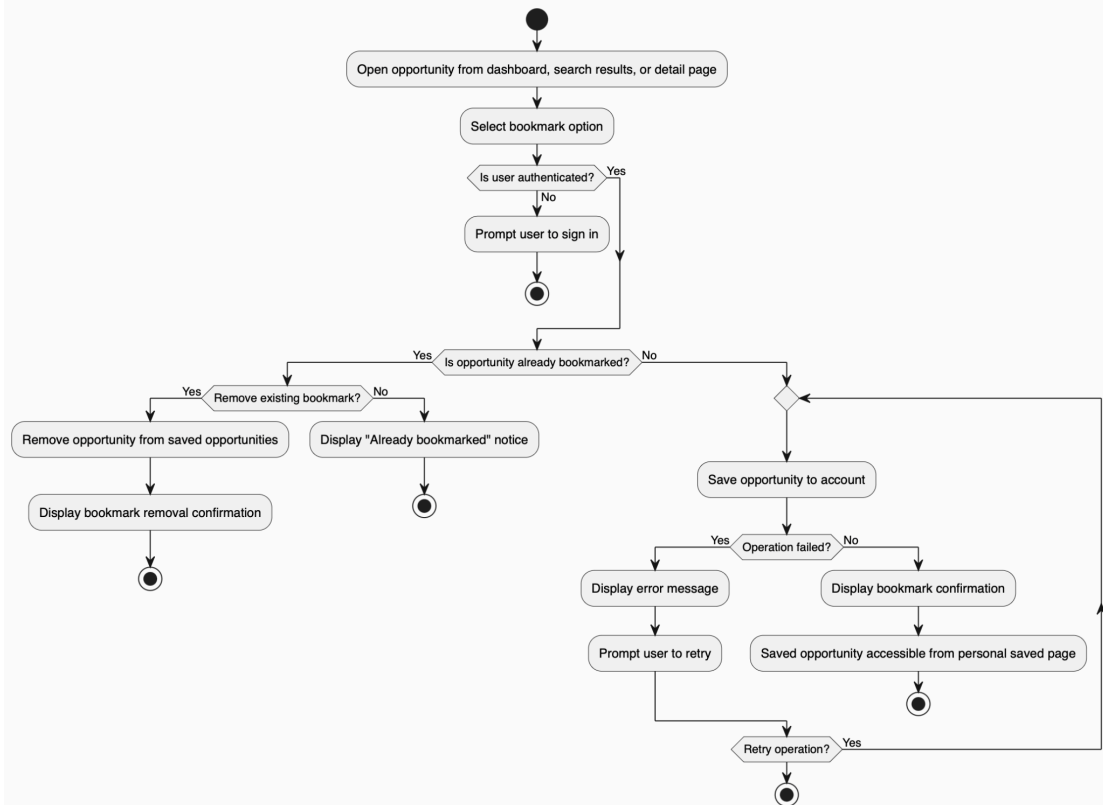
- If the user is not authenticated, the system prompts the user to sign in.
- If the opportunity is already bookmarked, the system does not create a duplicate bookmark.
- The user may remove an existing bookmark.
- If the operation fails, the system displays an error and allows the user to retry.

Postconditions:

- The bookmarked opportunity is stored in the user's account.
- The bookmark remains available across authenticated sessions.

User Flow Diagram (Bookmark Opportunity):

UC-4: Bookmark Opportunity



UC-5: Compare Opportunities

Use Case Type: Casual

Primary Actor: Registered User

Description:

A registered user compares two or more saved procurement opportunities side by side.

Preconditions:

- The user is authenticated.
- The user has bookmarked at least two opportunities.

Main Flow:

1. The user opens the saved-opportunities page.
2. The system displays the user's bookmarked opportunities.
3. The user selects two or more opportunities.
4. The user chooses the compare option.
5. The system displays a side-by-side comparison.
6. The comparison includes:
 - Agency
 - Budget
 - Submission deadline
 - Project category
 - Scope
 - Deliverables
 - Bidder qualifications
7. The user reviews the differences between the opportunities.

Alternative Flows:

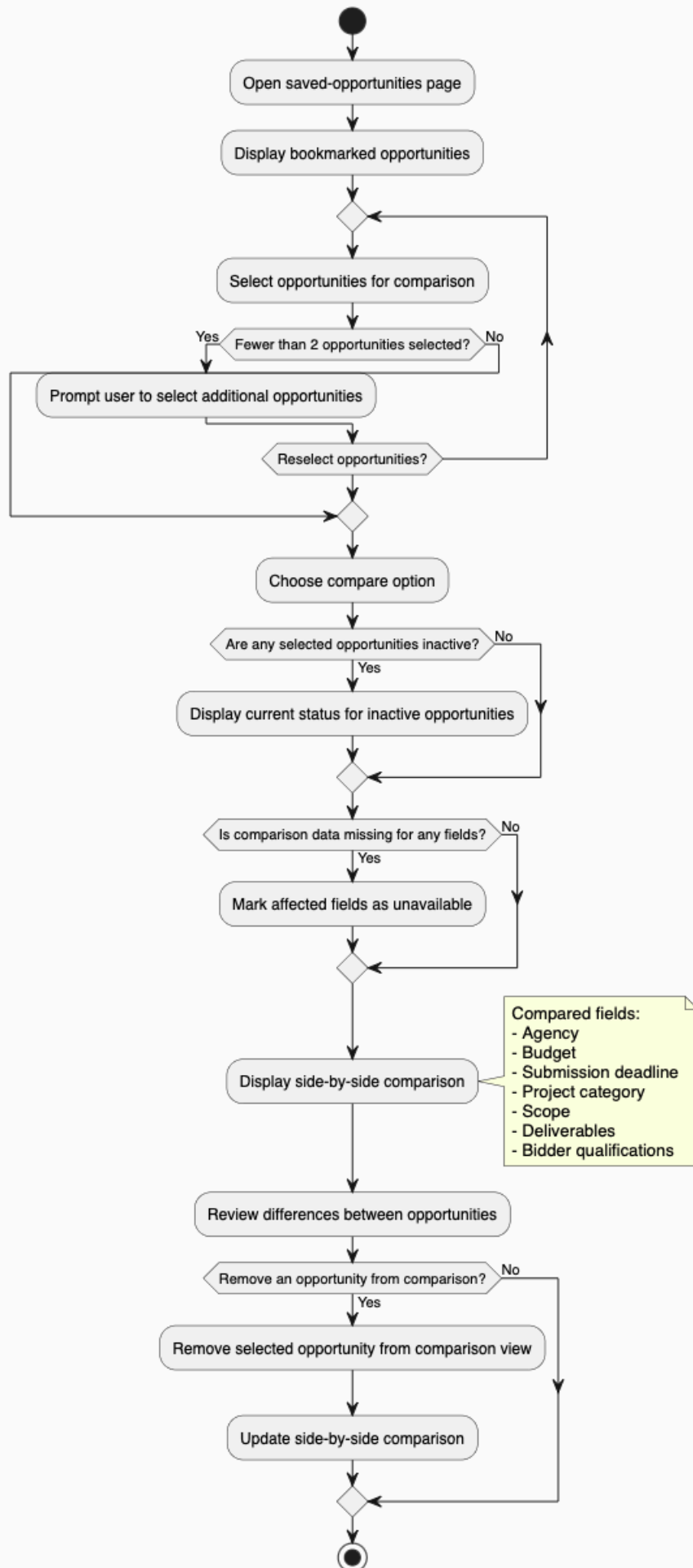
- If fewer than two opportunities are selected, the system asks the user to select additional opportunities.
- If an opportunity is no longer active, the system displays its current status.
- If comparison data is missing, the system marks the affected fields as unavailable.
- The user may remove an opportunity from the comparison.

Postconditions:

- The selected opportunities are displayed in comparison format.
- The underlying opportunity records remain unchanged.

User Flow Diagram (Compare Opportunities):

UC-5: Compare Opportunities



UC-6: View Historical Budget and Outlier Analysis

Use Case Type: Casual

Primary Actor: Registered User

Description:

The user reviews historical procurement budgets for similar projects and checks whether the selected opportunity's budget is unusually high or low.

Preconditions:

- The selected opportunity contains budget and category information.
- Historical comparable opportunities are available.

Main Flow:

1. The user opens an opportunity detail page.
2. The user selects the historical budget or price-analysis option.
3. The system identifies previous opportunities in the same or similar category.
4. The system retrieves historical budget information.
5. The system displays the information using charts or comparative statistics.
6. The system compares the selected opportunity's budget with historical records.
7. The system indicates whether the budget is within the normal range or is a statistical outlier.
8. The user reviews the result to support bid-planning decisions.

Alternative Flows:

- If insufficient historical data is available, the system informs the user that analysis cannot be completed.
- If the opportunity has no extracted budget, the system disables outlier analysis.
- If an outlier threshold has not been configured, the system displays only the historical comparison.
- If historical-data retrieval fails, the system displays an error and allows the user to retry.

Postconditions:

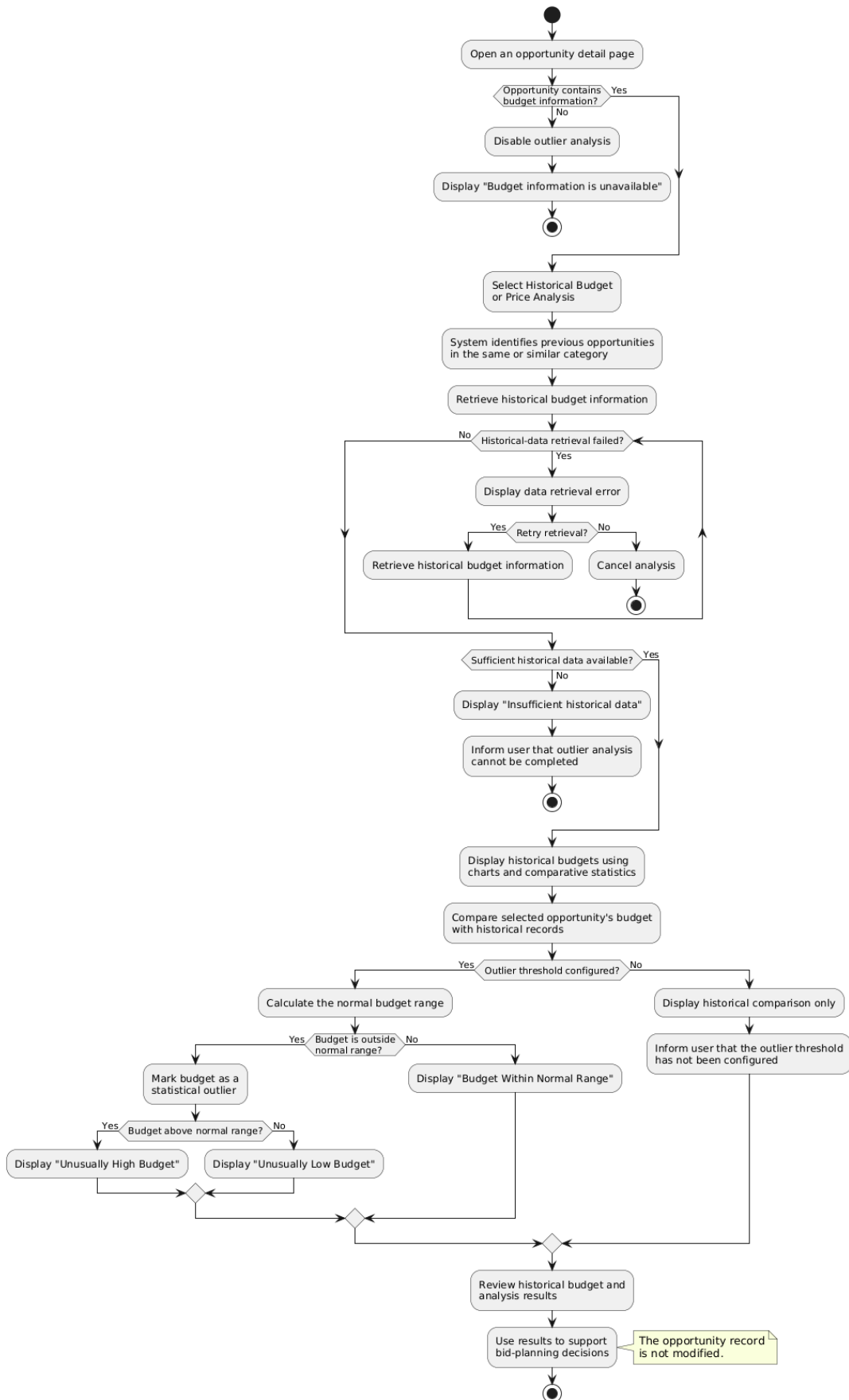
- Historical budget information is displayed.
- The opportunity record is not modified.

Note:

The exact statistical method and outlier threshold must be confirmed with the client.

User Flow Diagram (View Historical Budget and Outlier Analysis):

UC-6: View Historical Budget and Outlier Analysis



UC-7: Configure and Receive Opportunity Alerts

Use Case Type: Casual

Primary Actor: Registered User

Supporting Actors: Data Collection Service, Email Delivery Service

Description:

A registered user configures matching criteria and receives in-app or email notifications when new procurement opportunities match those criteria.

Preconditions:

- The user is authenticated.
- The user has a valid email address.

Main Flow:

1. The user opens notification preferences.
2. The system displays the available notification criteria.
3. The user configures one or more of the following:
 - Keywords
 - Software-project categories
 - Budget range
 - Issuing agencies
4. The user selects notification channels:
 - In-app notification
 - Email notification
 - Both channels
5. The user saves the preferences.
6. The Data Collection Service detects and classifies a new opportunity.
7. The system compares the opportunity with saved user preferences.
8. If a match is found, the system creates an in-app notification.
9. If email notification is enabled, the Email Delivery Service sends an email.
10. The user opens the notification and accesses the matching opportunity.

Alternative Flows:

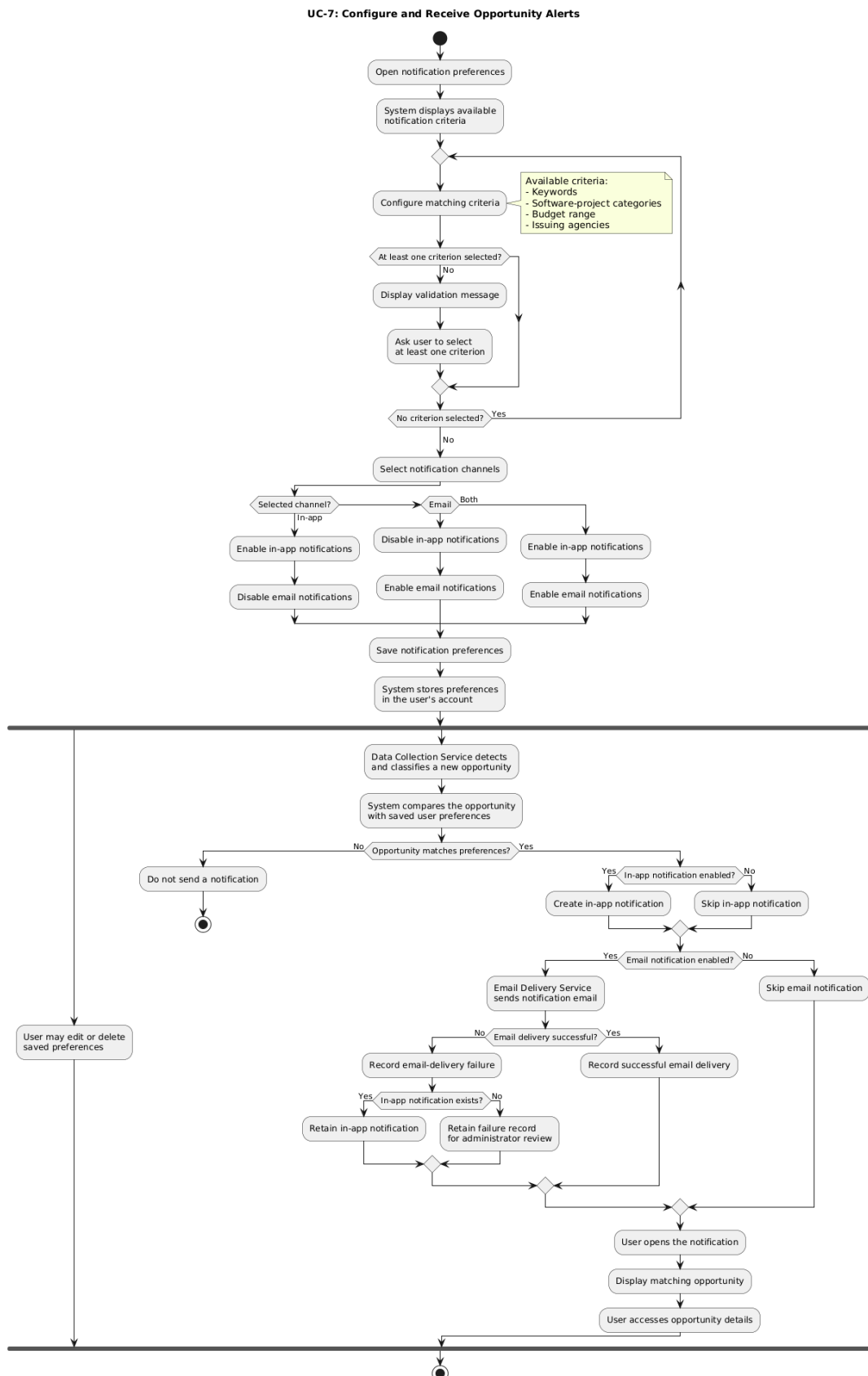
- If no new opportunity matches, no notification is sent.
- The user may disable email while keeping in-app notifications active.
- The user may disable in-app notifications while keeping email active.
- If email delivery fails, the system records the failure while retaining the in-app notification.
- The user may edit or delete saved notification preferences.

Postconditions:

- Notification preferences are stored in the user's account.

- The user receives notifications for future matching opportunities.

User Flow Diagram (Configure and Receive Opportunity Alerts):



UC-8: Collect Procurement Announcements

Use Case Type: Casual

Primary Actor: Data Collection Service

Supporting Actor: Government Procurement Data Source

Description:

The automated Data Collection Service checks monitored government websites for newly published procurement announcements and TOR documents.

Preconditions:

- At least one government procurement source is configured.
- The source is publicly accessible.
- The scheduled collection process is active.

Main Flow:

1. The Data Collection Service starts according to its configured schedule.
2. The service connects to each monitored government procurement source.
3. The service retrieves recently published announcements.
4. The service checks whether each announcement already exists in TORBIDD.
5. For a new announcement, the service captures available metadata.
6. The service locates and downloads the associated TOR document.
7. The service stores the original source information.
8. The downloaded document is submitted for extraction and classification.

Alternative Flows:

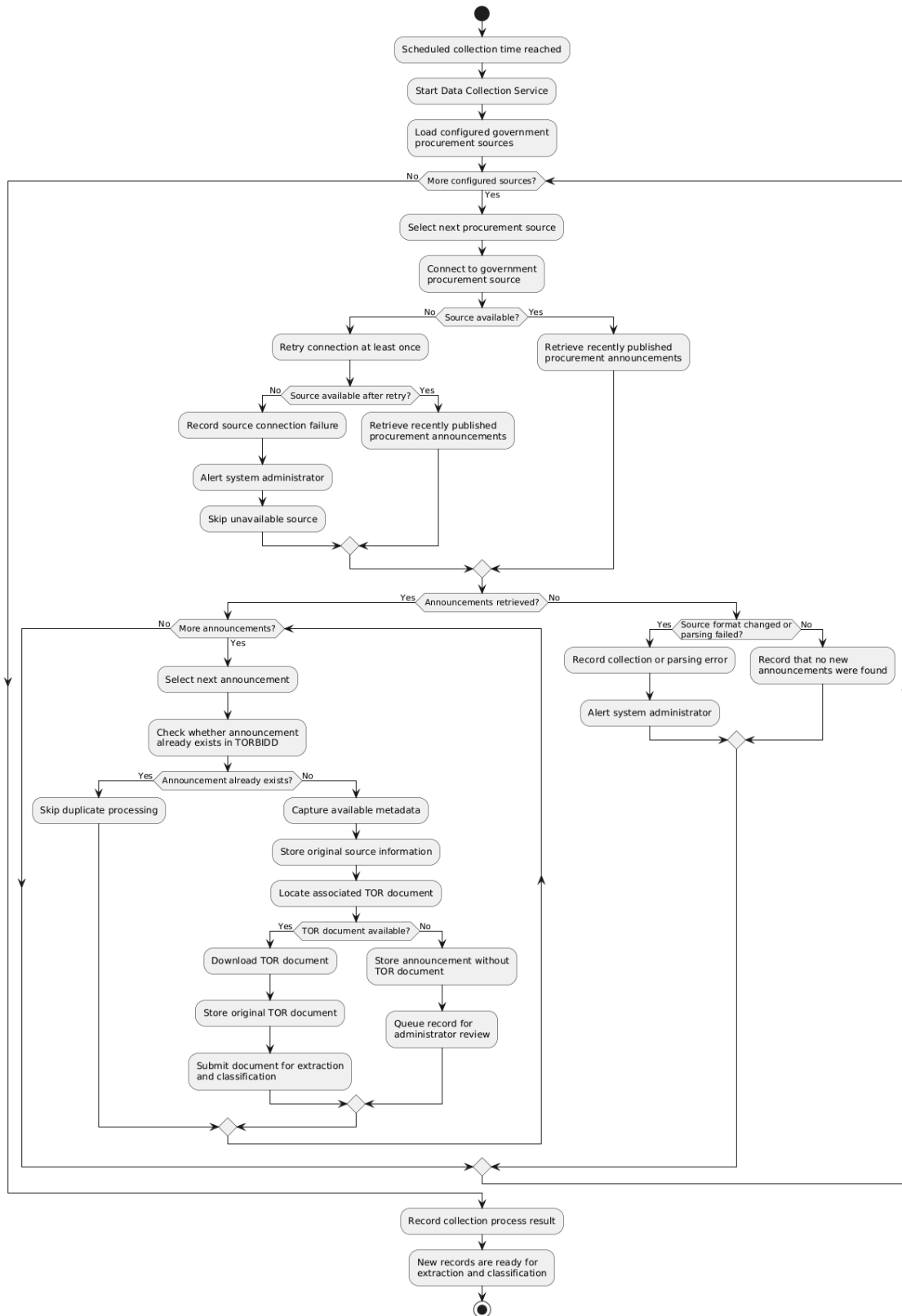
- If a source is unavailable, the service retries the request at least once.
- If the source remains unavailable, the system records the failure and alerts an administrator.
- If an announcement already exists, the system skips duplicate processing.
- If no TOR document is available, the record is queued for administrator review.
- If a source changes its format, the system records a collection or parsing error.

Postconditions:

- A new announcement and its available source document are stored.
- The record is ready for extraction and classification.

User Flow Diagram (Collect Procurement Announcements):

UC-8: Collect Procurement Announcements



UC-9: Extract and Classify TOR Information

Use Case Type: Casual

Primary Actor: Data Collection Service

Supporting Actor: Third-Party AI Service

Description:

The system processes a TOR document, extracts structured procurement fields, and classifies the listing as a software or non-software project.

Preconditions:

- A new procurement announcement and TOR document have been collected.
- The Third-Party AI Service is available.

Main Flow:

1. The Data Collection Service submits the TOR document to the AI service.
2. The AI service processes Thai document content.
3. The service extracts structured information, including:
 - Budget
 - Submission deadline
 - Project scope
 - Deliverables
 - Bidder qualifications
 - Contact information
4. Classification logic determines whether the listing is a software project.
5. If it is software-related, the system assigns a project category.
6. The system records extraction and classification confidence indicators.
7. The system validates the extracted fields.
8. If confidence is acceptable, the structured record is published to the opportunity dashboard.
9. If confidence is low, the record is added to the administrator review queue.

Alternative Flows:

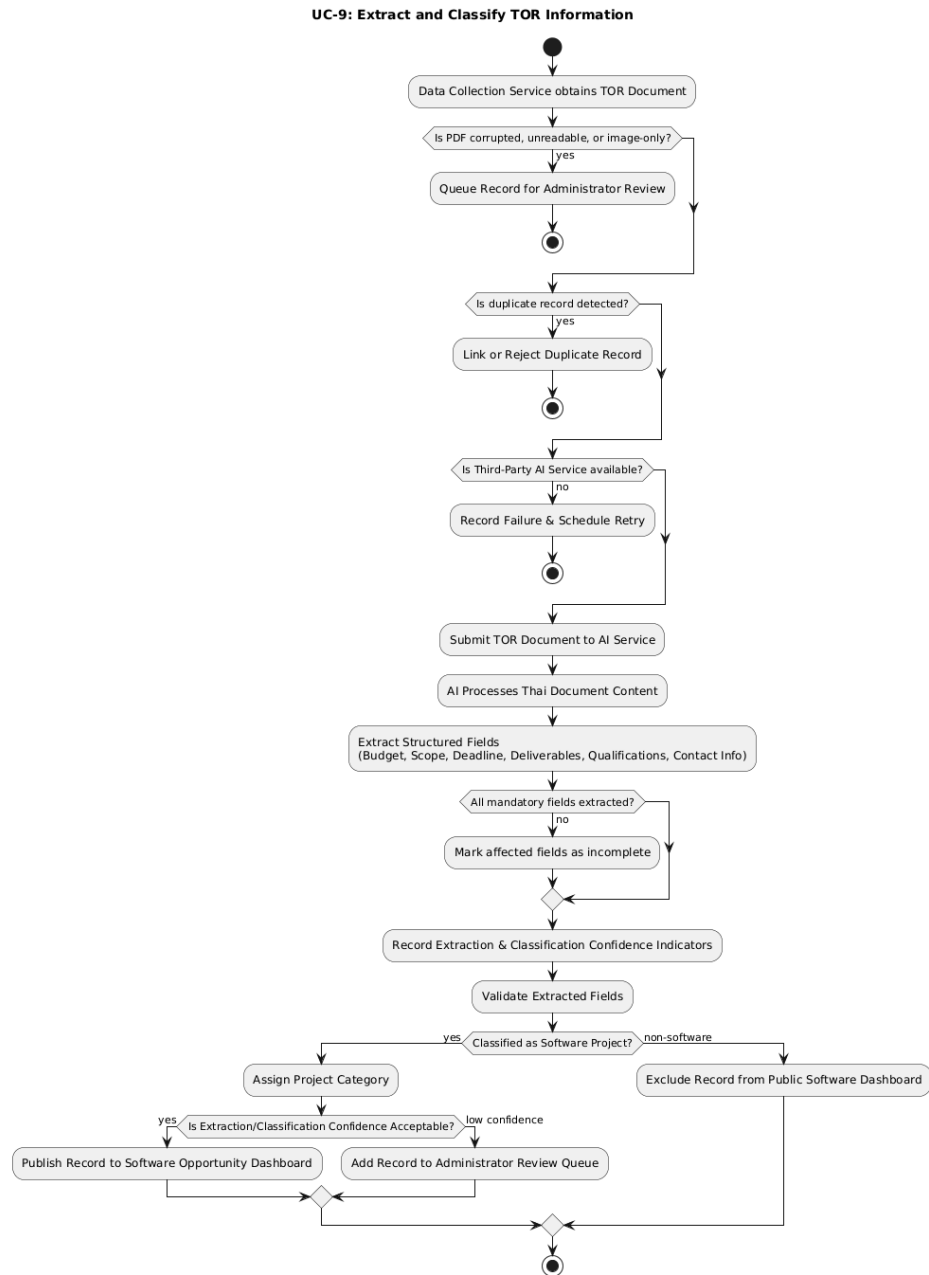
- If the PDF is corrupted, unreadable, or image-only, the record is queued for administrator review.
- If the AI service is unavailable, the system records the failure and schedules a retry.
- If mandatory fields cannot be extracted, the system marks the affected fields as incomplete.
- If the listing is classified as non-software, the system prevents it from appearing on the public software-opportunity dashboard.
- If a duplicate is detected, the system links or rejects the duplicate record.

Postconditions:

- A structured opportunity record is created.

- The record is either published, excluded as non-software, or queued for review.

User Flow Diagram (Extract and Classify TOR Information):



UC-10: Review and Correct Opportunity Information

Use Case Type: Casual

Primary Actor: System Administrator

Description:

An administrator reviews low-confidence or failed extraction records and corrects their fields or project classification.

Preconditions:

- The administrator is authenticated.
- At least one record is available in the review queue.

Main Flow:

1. The administrator opens the administration panel.
2. The administrator accesses the extraction and classification review queue.
3. The system displays flagged records with confidence indicators.
4. The administrator selects a record.
5. The system displays the extracted fields and original TOR document.
6. The administrator compares the structured information with the source document.
7. The administrator edits incorrect or incomplete fields.
8. The administrator changes the software or non-software classification if necessary.
9. The administrator approves the corrected record.
10. The system records the administrator's changes.
11. If the record is software-related, the system publishes it to the dashboard.

Alternative Flows:

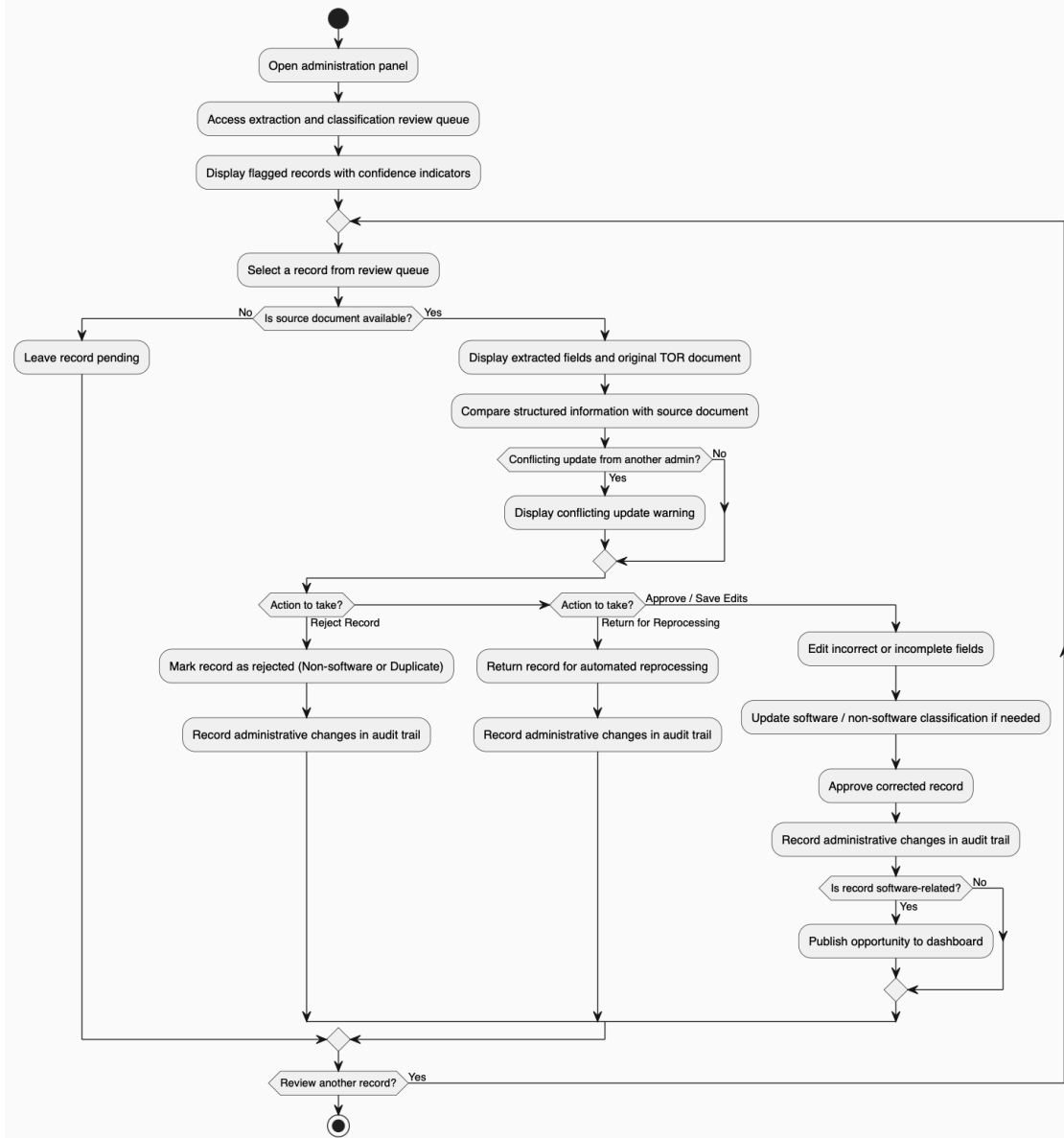
- The administrator rejects a non-software record.
- The administrator rejects a duplicate record.
- If the source document is unavailable, the administrator leaves the record pending.
- The administrator may return the record for automated reprocessing.
- If another administrator has modified the record, the system warns about the conflicting update.

Postconditions:

- The record contains corrected information.
- The record is published, rejected, or remains pending.
- The system retains an audit trail of administrative changes.

User Flow Diagram (Review and Correct Opportunity Information):

UC-10: Review and Correct Opportunity Information



UC-11: Manage Users

Use Case Type: Casual

Primary Actor: System Administrator

Description:

An administrator manages TORBIDD user accounts and access roles.

Preconditions:

- The administrator is authenticated.
- The administrator has user-management permission.

Main Flow:

1. The administrator opens the user-management page.
2. The system displays registered user accounts.
3. The administrator searches for or selects a user.
4. The system displays the account's basic information, role, and status.
5. The administrator performs an authorized action, such as:
 - Change the user's role
 - Activate an account
 - Suspend an account
 - Restore a suspended account
6. The system validates the change.
7. The system updates the user account.
8. The system records the administrative action.

Alternative Flows:

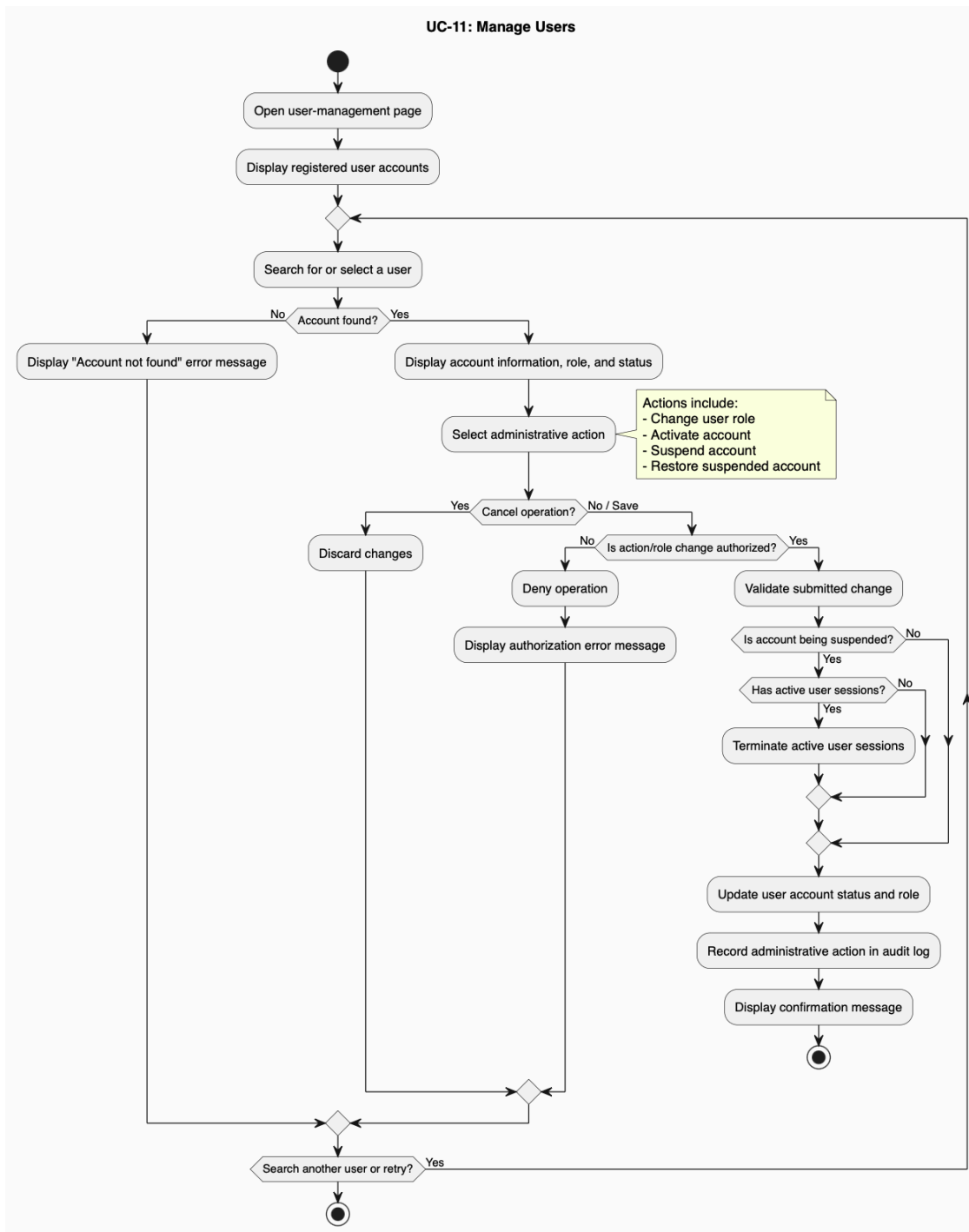
- If the administrator attempts an unauthorized role change, the system denies the operation.
- If the account cannot be found, the system displays an error.
- If the account has active sessions when suspended, the system terminates those sessions.
- The administrator may cancel before saving changes.

Postconditions:

- The selected user account reflects the updated role or status.
- The administrative action is recorded.

User Flow Diagram (Manage Users):

UC-11: Manage Users



UC-12: Manage User Profile and Preferences

Primary Actor: Registered User

Description:

A registered user views and updates account information and notification preferences.

Preconditions:

- The user is authenticated.

Main Flow:

1. The user opens the profile page.
2. The system displays the user's account information.
3. The user edits permitted fields such as:
 - Name
 - Organization
 - Organization or professional role
 - Notification preferences
4. The user saves the changes.
5. The system validates the submitted information.
6. The system updates the user profile.
7. The system displays confirmation.

Alternative Flows:

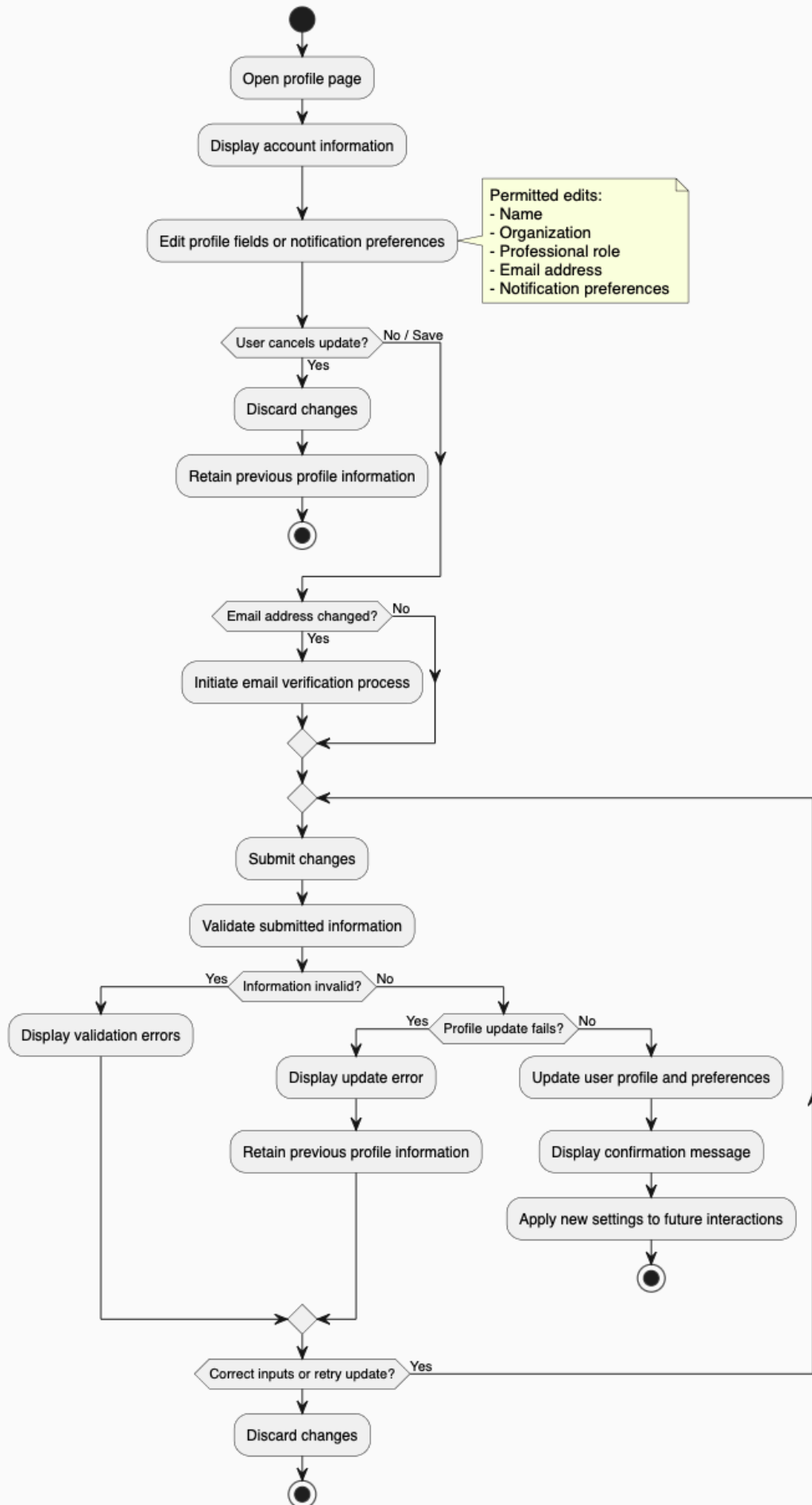
- If information is invalid, the system displays validation errors.
- If the email address is changed, the system may require email verification.
- The user may cancel without saving.
- If the update fails, the system retains the previous information.

Postconditions:

- The user profile and preferences are updated.
- The new settings apply to future system interactions and notifications.

User Flow Diagram (Manage User Profile and Preferences):

UC-12: Manage User Profile and Preferences



6. Nonfunctional Requirements

This section specifies the quality attributes, performance expectations, security requirements, and operational constraints of TORBIDD.

6.1 Performance Requirements

- **NFR-6.1.1** Dashboard search and filter results shall be returned within 3 seconds under normal operating conditions. **(High)**
- **NFR-6.1.2** Newly published procurement announcements shall be collected and made available on the dashboard within 24 hours of publication, subject to source availability. **(Medium)**
- **NFR-6.1.3** Procurement detail pages shall load within 3 seconds under normal operating conditions, excluding the time required to retrieve external documents. **(Medium)**

6.2 Security Requirements

- **NFR-6.2.1** All client-server communication shall use HTTPS/TLS. **(High)**
- **NFR-6.2.2** The system shall restrict administrative functions to authenticated administrator accounts. **(High)**
- **NFR-6.2.3** Users shall only be able to access and modify their own personal data and save procurement opportunities. **(High)**
- **NFR-6.2.4** The system shall use appropriate authentication and authorization mechanisms to prevent unauthorized access to protected functions and data. **(High)**

6.3 Usability Requirements

- **NFR-6.3.1** The dashboard, search, and procurement detail views shall be usable on desktop and mobile screen sizes. **(High)**
- **NFR-6.3.2** Procurement information extracted from TOR documents shall be presented in a clear and structured format. **(High)**
- **NFR-6.3.3** The system interface shall support Thai content where applicable. **(Medium)**
- **NFR-6.3.4** Error and status messages shall be presented in clear and understandable language. **(Medium)**

6.4 Reliability Requirements

- **NFR-6.4.1** The data collection service shall retry a failed source request at least once before marking the source as requiring administrator attention. **(Medium)**
- **NFR-6.4.2** The dashboard shall remain available using previously stored procurement data if the AI extraction service is temporarily unavailable. **(High)**
- **NFR-6.4.3** Temporary failure of one government data source shall not prevent the system from processing data from other available sources. **(Medium)**

6.5 Maintainability Requirements

- **NFR-6.5.1** Data collection logic for each government source shall be modular so that sources can be added or modified without significantly affecting unrelated collection components. **(Medium)**
- **NFR-6.5.2** The system shall use a modular architecture that allows AI processing, data collection, search, and notification components to be modified independently where practical. **(Medium)**

6.6 Portability Requirements

- **NFR-6.6.1** The application shall be deployable using Docker on a compatible cloud-hosting environment. **(High)**
- **NFR-6.6.2** The web application shall support modern versions of commonly used browsers, including Google Chrome, Mozilla Firefox and Safari. **(Medium)**

6.7 Scalability Requirements

- **NFR-6.7.1** The database and application architecture shall support the addition of further government agencies without requiring fundamental redesign of the system. **(Medium)**
- **NFR-6.7.2** The system shall support growth in the number of procurement records and users without requiring fundamental changes to the core architecture. **(Medium)**

6.8 Other Non-Functional Constraints

- **NFR-6.8.1** The system shall support Thai-language procurement documents and extracted content, as government TOR documents may be written in Thai.
- **NFR-6.8.2** The system shall depend on publicly available government data sources and third-party services described in Sections 2.6 and 3.3.

6.9 Business Rules

The following business constraints apply to the project:

- **BUS-6.9.1** The total project budget shall not exceed ฿1,000,000.
- **BUS-6.9.2** The planned project delivery period shall be 10 weeks from requirements sign-off.
- **BUS-6.9.3** TORBIDD shall operate as an information aggregation and intelligence platform and shall not submit bids or modify procurement information on behalf of users.
- **BUS-6.9.4** Procurement information presented by TORBIDD shall originate from publicly available government sources.

Budget Allocation

Project / Initiative	Value (฿)	% of Total
Procurement Opportunity Dashboard	130,960	13.1%
Smart Search & Filtering	109,780	11.0%

TOR Information Extraction	154,960	15.5%
Historical Price Dashboard	88,590	8.9%
Opportunity Notification	45,600	4.6%
Testing, Deployment & Contingency	48,920	4.9%
Core Features	421,190	42.1%
Total Project Budget	1,000,000	100%

7. Other Requirements

7.1 Legal and Compliance Requirements

- **LEGAL-1:** The system shall use procurement information obtained from publicly available government sources and shall not modify or submit procurement information to government procurement systems.
- **LEGAL-2:** The system shall handle bidder, agency, and contact information in accordance with applicable data-protection and privacy requirements when such information is collected or stored.
- **LEGAL-3:** The system shall not intentionally collect or store personally identifiable bidder information beyond what is necessary for the system's intended procurement-information functions.
- **LEGAL-4:** The system shall comply with the applicable terms and usage policies of third-party services used by the system, including Vertex AI and external data sources.

7.2 Localization Requirements

- **LOC-1:** The system shall support **Thai procurement content**, including TOR documents and extracted information.
- **LOC-2:** User-interface language requirements shall be confirmed during detailed UI design. At minimum, the system shall correctly display Thai text.

7.3 Out-of-Scope Features

The following features are outside the scope of the current TORBIDD implementation:

- Submission of bids or procurement documents on behalf of users
- Modification of information on government procurement systems
- Direct authentication with government procurement systems
- Private or restricted procurement data that is not publicly available
- Automated bidding or bid decision-making
- Payment processing or monetization features
- Native mobile applications
- Advanced AI model training or active learning from administrator corrections
- Integration with additional government sources beyond the agreed initial scope without further development

7.4 Development and Testing

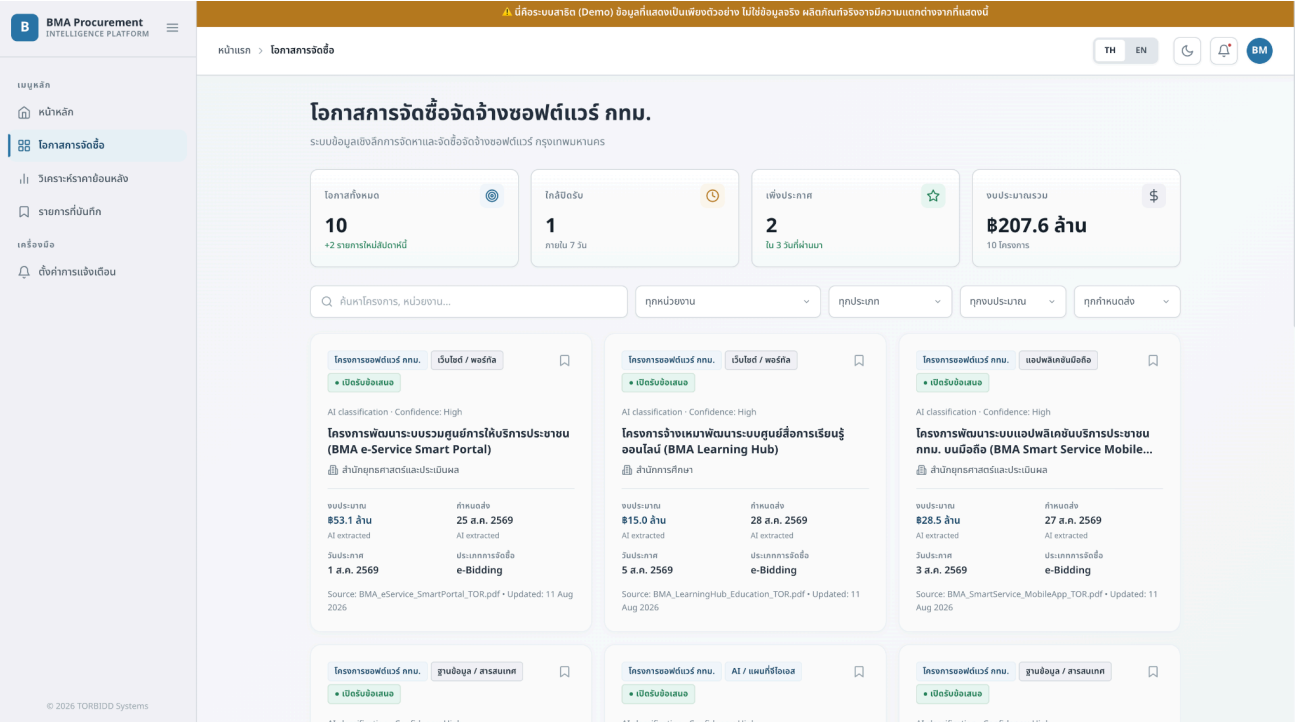
- **DEV-1:** Testing and validation shall be performed by the project team throughout development rather than through a dedicated QA role.
- **DEV-2:** The system shall be validated against the functional and non-functional requirements specified in this SRS before final delivery.

8. Additional Material

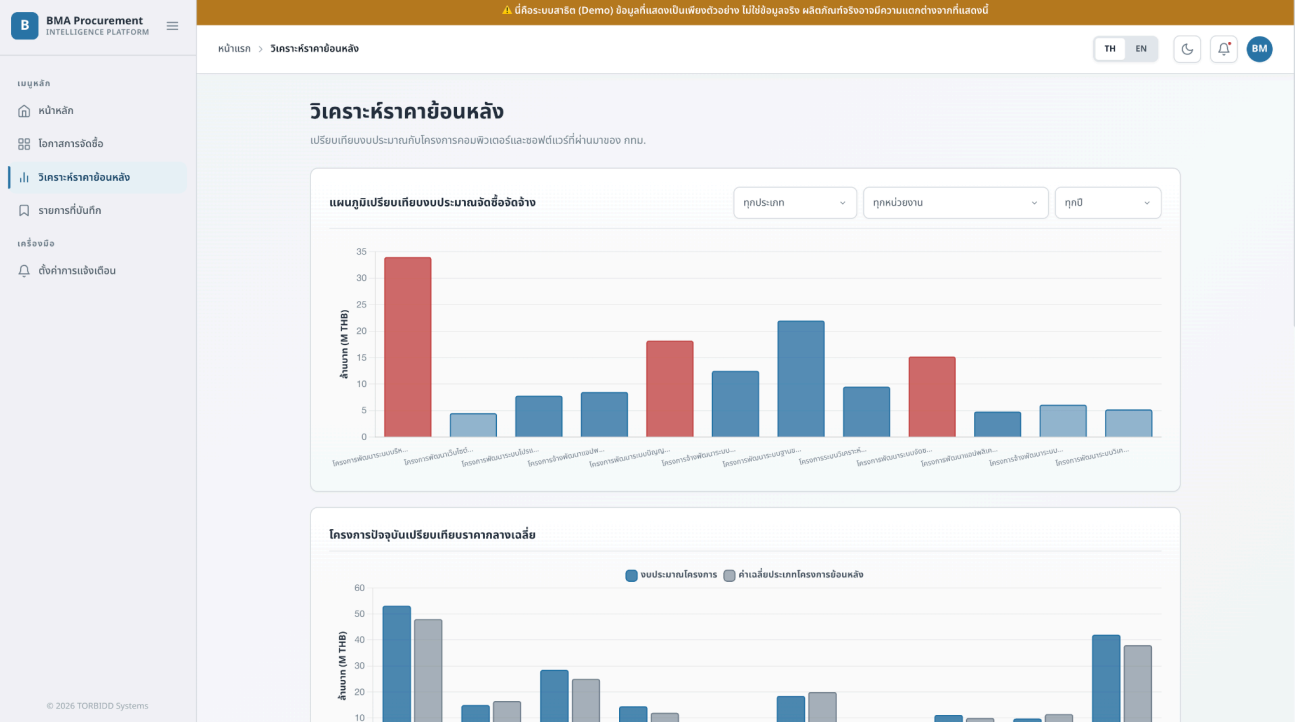
8.1 Website Preview



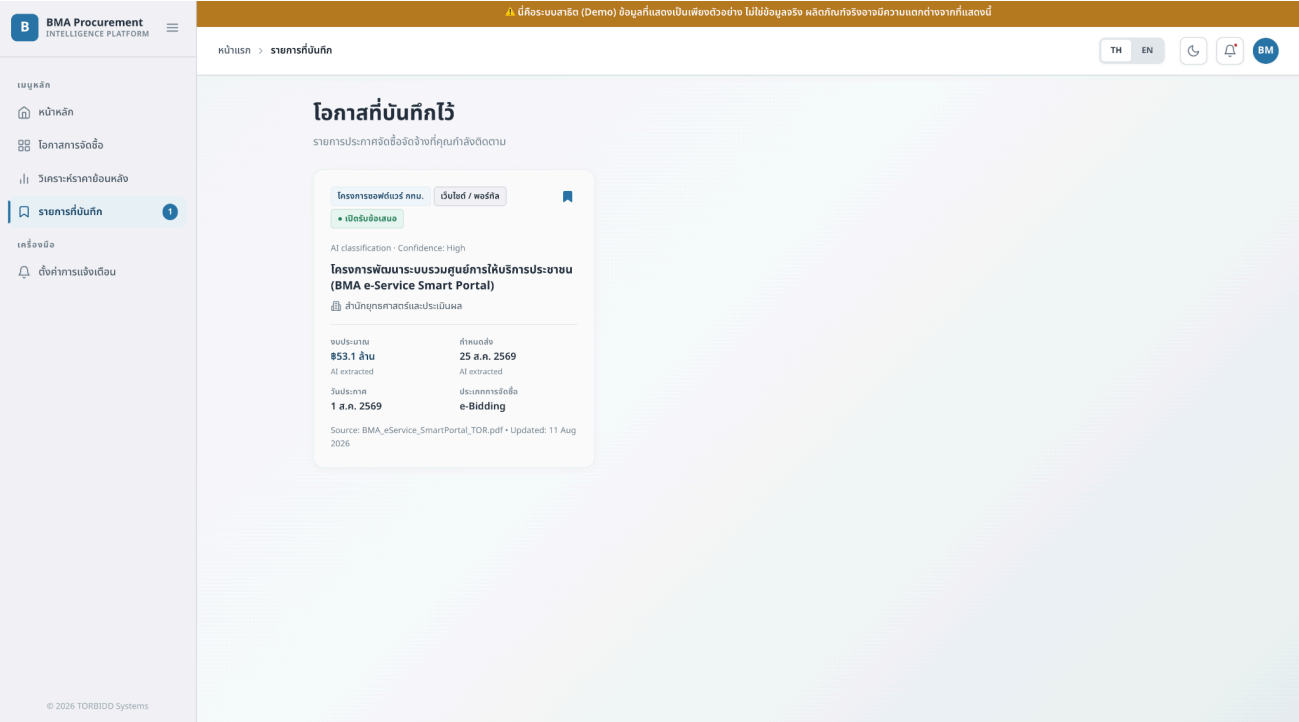
Home page



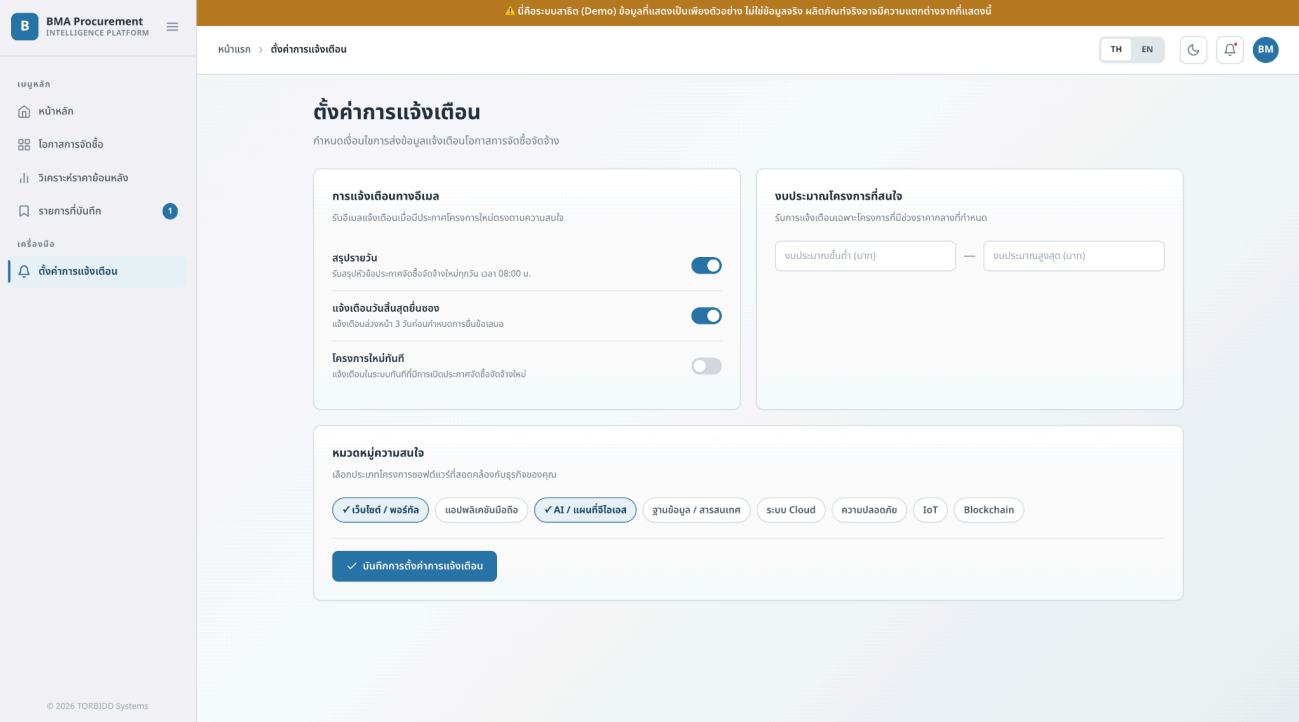
Opportunities board page



Price Analysis page



Saved Items page



Notification Setting Page

Appendix A: Glossary

Term	Definition
TOR	Terms of Reference — the official document an agency publishes describing a procurement's scope, budget, deliverables, and bidder qualifications.
BMA	Bangkok Metropolitan Administration, the primary initial data source for procurement announcements.
TORBIDD	The name of the platform being specified in this document.
Extraction confidence	A system-generated indicator of how reliable an AI-extracted field is, used to route low-confidence records to administrator review.
Outlier analysis	Statistical comparison flagging an opportunity's budget as unusually high or low relative to similar historical opportunities.
Vertex AI	The AI platform is referenced in the proposal as the intended engine for TOR PDF extraction and classification.
MongoDB Atlas	The cloud-hosted database referenced in the proposal for storing structured opportunity data.